CLINIC APPOINTMENT AND PATIENT MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

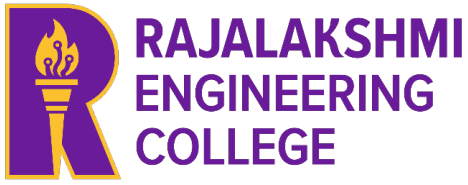
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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
Computer Science & Business System



RAJALAKSHMI ENGINEERING COLLEGE
(An Autonomous Institution, Affiliated to Anna University, Chennai)
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2026-2027

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BONAFIDE CERTIFICATE

Certified that this project report titled “Clinic Appointment and Patient Management System” is the bonafide work of **RENE CHRISTINE A (241401079)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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I, **RENE CHRISTINE A (241401079)**, hereby declare that the project report titled "Clinic Appointment and Patient Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: _____

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ABSTRACT

Small and medium-sized clinics still run a large part of their daily work on paper registers and loosely maintained spreadsheets. Patient details, appointment slots, consultation notes and bills are kept in separate places, so receptionists cannot easily see whether a doctor is free, doctors cannot quickly retrieve a patient's history, and the management has no consolidated view of how many patients were seen or how much is still unpaid. Double bookings, lost records and delayed billing are common consequences.

This project presents the Clinic Appointment and Patient Management System, a web and mobile application developed on the Zoho Creator low-code platform. The application is organised into the sections Patients, Doctors, Appointments, Treatments and Billing, supported by an App Preferences section and a Clinic Dashboard page. Five relational forms capture the clinic's master and transaction data; lookup fields connect each appointment to a patient and a doctor, each treatment to an appointment, and each bill to a patient and an appointment, so that a patient's complete visit history can be traced from a single record. An Upcoming Appointments report gives the front desk a live list of scheduled visits, and the dashboard summarises clinic activity. A role-based access model for the Administrator, Receptionist and Doctor roles has been designed, and a Deluge script, run through a Zoho Creator connection, creates a Zoho Calendar event for every appointment.

Using the low-code approach, the data model, forms and reports were assembled in a fraction of the time a conventional web stack would need, while hosting, security and mobile access were provided by the platform. The application replaces manual registers with a single searchable source of truth, reduces duplicate data entry and gives the clinic immediate visibility of its schedule, treatments and pending payments.

Sample-data testing confirmed duplicate-phone validation and the main appointment, billing and dashboard views. Stage and Production list version 2.0 (01 November 2026). Some test cases, role-based sign-in, calendar-event execution and user feedback remain unverified; no performance improvement is claimed.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Clinic Management; Appointment Scheduling; Role-Based Access Control

PROJECT AT A GLANCE

Project Title	Clinic Appointment and Patient Management System
Application Domain	Healthcare – outpatient clinic management
Student Name & Register No.	Rene Christine A – 241401079
Zoho Creator Edition	Trial edition (trial banner shown during development)
Application Name	Clinic Appointment & Patient Management System
Published / Live URL	https://creatorapp.zoho.in/renechristine.a.2024.csbs324/clinic-appointment-patient-management-system
Demo Credentials (per role)	Roles available for the demonstration: Admin, Doctor, Receptionist and Accountant (CEO at the top of the hierarchy). User names and passwords are not printed in this report and are provided separately at the demonstration.
Number of Forms / Subforms	5 / 0 (Patients, Doctors, Appointments, Treatments and Billing)
Number of Reports (menu items)	10 report views: All Patients, All Doctors, All Appointments (2 views), Today's Appointments, Upcoming Appointments, Appointment Calendar, All Treatments, All Billings and Pending Bills.
Number of Pages / Dashboards	1 (Clinic Dashboard)
Number of Workflows	4 form workflows and 1 scheduled workflow (Appointment Reminder Scheduler)
Deluge Functions / Custom Actions	1 Deluge script (DS-01: Zoho Calendar event creation)
Integrations	Zoho Calendar through the Creator connection "zoho_calendar_connection".
Roles & Profiles Defined	CEO, Admin, Doctor, Receptionist and Accountant; custom permission sets are defined for Admin, Doctor, Receptionist and Accountant.
Environments Used	Development → Stage 2.0 → Production 2.0
Date of Production Deployment	01 November 2026 (version 2.0 is listed in Stage and Production).

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
OPD	Outpatient Department
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Outpatient clinics are the first point of contact between patients and the healthcare system. A typical small clinic in an Indian city has a few consulting doctors, one or two front-desk staff and a daily flow of between thirty and a hundred patients. Each visit passes through the same sequence: the patient is registered or recognised, an appointment is fixed with a doctor, the consultation takes place, a diagnosis and prescription are written, and a bill is raised and collected.

In many such clinics this sequence is still handled with a paper register, appointment diaries and handwritten bills, sometimes supplemented by a spreadsheet. The people involved are the receptionist, who books and tracks appointments and collects payments; the doctors, who need the patient's history and who record the diagnosis and prescription; and the clinic owner or administrator, who needs to know how the clinic is performing. This project builds a Zoho Creator application that brings all of these activities into one connected system.

1.2 Problem Statement

The clinic has no single, reliable record that links a patient to their appointments, treatments and payments. Information is scattered across registers and files, which makes scheduling error-prone, slows down consultations and makes it difficult for the management to monitor the clinic. A centralised, easy-to-use application is needed that can be built and maintained without a large software team.

- Patient details are written again at every visit, causing duplicate and inconsistent records.
- Appointment slots are tracked in diaries, so double bookings and missed appointments occur.
- Previous diagnoses and prescriptions cannot be retrieved quickly during a consultation.
- Bills are prepared manually and unpaid bills are noticed late.
- Month-end reports have to be compiled by hand, and there is no role-wise control of who can see patient data.

1.3 Objectives

- To design a normalised data model and forms for patients, doctors, appointments, treatments and billing.
- To automate appointment handling using Deluge scripting, including creating a Zoho Calendar event for every appointment.
- To provide reports and a dashboard that show upcoming appointments and clinic activity at a glance.
- To restrict access to patient data through role-based permissions for the Administrator, Receptionist and Doctor.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

In scope: registration and maintenance of patients and doctors; booking and status tracking of appointments; recording of diagnosis, prescription and treatment notes; billing with payment status; reports and a dashboard; and role-based access for three user roles. The application is accessed through a web browser and through the Zoho Creator mobile app.

Out of scope: online payment gateways, insurance claim processing, laboratory and pharmacy inventory management, video consultation, and a self-service patient portal. Some of these are listed as future enhancements.

1.5 Justification for a Low-Code Approach

A clinic of this size rarely has an in-house development team, and a conventional web application would need a database, a back-end, a front-end, hosting and security to be designed and maintained separately [1]. A low-code application platform such as Zoho Creator supplies these layers as managed services: forms generate the database tables and data-entry screens, reports and pages are assembled visually, and Deluge scripting is used only where custom logic is needed [2]. This shortens development time, allows a domain user to maintain the application, provides mobile access without a separate app, and removes hosting and infrastructure cost for the clinic.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirements, and Chapter 4 describes the system design. Chapter 5 explains the Zoho Creator implementation. Chapter 6 discusses testing and debugging; Chapter 7 covers deployment, security and governance; Chapter 8 presents the results; and Chapter 9 concludes with future enhancements. References and appendices follow.

How the chapters connect

The report follows the development of the application from the problem definition to the evidence available at the end of implementation. The requirement analysis in Chapter 3 establishes what the clinic roles need to do; the design in Chapter 4 translates those needs into linked forms, reports, workflows and access rules. Chapter 5 then records how those elements were configured in Zoho Creator. This sequence makes it possible to trace an application feature back to a requirement and forward to the relevant test or output.

The later chapters distinguish implemented behaviour from planned work. Chapter 6 records the tests that were executed, the screenshots and record views used as evidence, and the defects that remain open. Chapter 7 describes deployment and security configuration, while Chapter 8 discusses the observed outputs and the limitations of the sample-data evaluation. The conclusion therefore treats unverified role sign-in, calendar execution and user acceptance as future verification tasks rather than as completed results.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system, a patient arrives or telephones the clinic and the receptionist writes the name and requested time in a paper register or diary. Availability of a doctor is judged from memory or by scanning the diary. After the consultation the doctor notes the diagnosis and prescription on a paper card kept in a file, and the bill is written by hand in a separate book. At month end the staff count entries manually to prepare reports for the owner. The flow is shown in Figure 2.1.

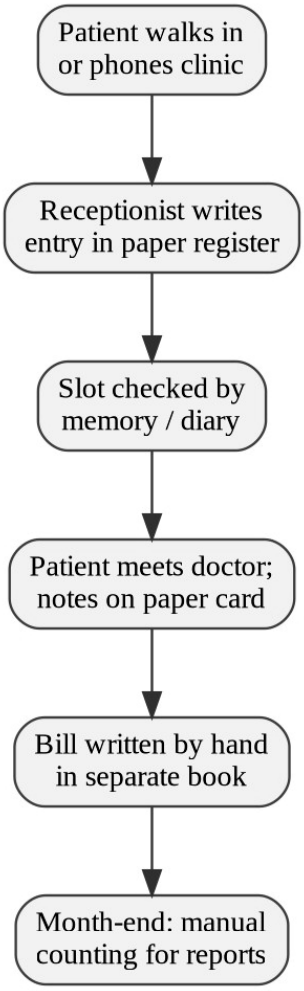


Figure 2.1: As-Is Process Flow of the Existing Manual Clinic Process

2.2 Limitations of the Existing System

- Patient information is duplicated and cannot be searched quickly.
- Double bookings are not detected automatically and reminders are not sent.

- Treatment history is physically stored and easily misplaced.
- Billing and payment status are not linked to the consultation record.
- There is no real-time visibility of the schedule, revenue or pending payments, and no access control over patient data.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for this particular application. Published studies describe low-code platforms as a way to shorten development and to enable people with limited programming experience to build business applications [3]; the figures below are the author's own estimates for this project.

Table 2.1: Comparison of Low-Code and Traditional Development for the Clinic Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	6–8 weeks (estimated)	1–2 weeks (estimated)
Skills required	Front-end, back-end and database developers	Data modelling and basic Deluge scripting
Hosting & infrastructure	Server, database and deployment set-up by the clinic	Provided by the Zoho cloud
Mobile application	Separate Android/iOS development	Available through the Zoho Creator mobile app
Security & access control	Custom authentication and permission code	Built-in roles, profiles and permission sets
Maintenance & change	Code changes, testing and redeployment	Edit forms and reports visually and publish
Customisation limits	Unlimited	Limited by platform capabilities; Deluge covers most needs
Cost	Developer and server cost	Platform subscription only

2.4 Literature Survey

Table 2.2 summarises the sources reviewed. Their relevance to the present work is that they describe what low-code platforms can and cannot do, and they document the product capabilities used in this application.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide	Explains low-code concepts, platform capabilities and development practice.	Foundation for the low-code approach adopted.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
2	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, roles and deployment on Zoho Creator.	Primary technical reference for implementation.
3	R. Sanchis et al. [3], 2020	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Discusses low-code as a means of speeding up application development by non-specialists.	Supports the choice of low-code for a small organisation.
4	Y. Luo et al. [4], 2021	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Analyses practitioner discussions to identify popular uses and recurring challenges of low-code platforms.	Informs the limitations discussed later in the report.
5	A. C. Bock and U. Frank [5], 2021	Low-Code Platform	Gives a conceptual overview of low-code platforms, their characteristics, benefits and risks.	Context for platform selection and trade-offs.
6	Gartner, Inc. [6]	Magic Quadrant for Enterprise Low-Code Application Platforms	Evaluates vendors of enterprise low-code platforms as a distinct product category.	Context for platform selection.
7	Zoho Corporation [7]	Zoho Creator Learning Centre	Tutorials and guided exercises on building Creator applications.	Used for form and report configuration.

2.5 Emerging Trends in Low-Code Development

Industry analysts evaluate low-code platforms as a distinct product category [6], and recent platforms increasingly add AI assistance for generating applications, fields and scripts, a growing citizen-developer community, and workflow-driven hyper-automation. For this application the relevant trends are AI-assisted application building, which speeds up form creation, and the use of automation to replace manual follow-up such as reminders. Any AI-generated element used in the project is verified manually before use and is acknowledged in the report.

2.6 Proposed System

The proposed system is a Zoho Creator application with five connected forms. Patient and doctor records are entered once and reused through lookup fields, which removes duplicate entry. Appointments are stored with date, time and status so the schedule can be viewed in reports and on the dashboard; treatments and bills are linked to the appointment so a patient's visit history is available in one place. Role-based permissions protect patient data, and automation reduces manual follow-up. In this way each limitation listed in Section 2.2 is addressed by a specific feature of the application.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Clinic owner or manager with full control of the application.	Manage users and settings, maintain doctor records, view all data, reports and the dashboard.
Receptionist	Front-desk staff who deal with patients directly.	Register patients, book and update appointments, prepare bills and record payments.
Doctor	Consulting physician.	View own appointments, record diagnosis, prescription and treatment notes, update appointment status.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow staff to register a patient with name, date of birth, gender, phone number, e-mail, address, blood group and medical history.	Admin, Receptionist	High
FR-02	The system shall maintain doctor profiles with specialization, contact details and availability status.	Admin	High
FR-03	The system shall allow an appointment to be booked by selecting a patient and a doctor and entering date, time and reason for visit.	Receptionist	High
FR-04	The system shall maintain an appointment status (for example Scheduled) and allow it to be updated.	Receptionist, Doctor	High
FR-05	The system shall store the appointment time of each visit so that a doctor’s schedule can be seen day-wise.	All roles	Medium
FR-06	The system shall create a 60-minute event in Zoho Calendar for each appointment, using the appointment date and time.	System	Medium
FR-07	The system shall record diagnosis, prescription and treatment notes against an appointment.	Doctor	High
FR-08	The system shall record consultation fee, medicine charges and total amount in a bill linked to the patient and appointment.	Receptionist	High
FR-09	The system shall record the payment status of each bill.	Receptionist	Medium
FR-10	The system shall provide a list of upcoming appointments.	All roles	High
FR-11	The system shall provide a dashboard summarising clinic activity.	Admin, Doctor	Medium
FR-12	The system shall restrict access to forms and reports according to the user’s role.	Admin	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 1,000 records shall load within a few seconds.
NFR-02	Security	Users shall access only the data permitted by their role.
NFR-03	Usability	The application shall be usable on a web browser and on the Zoho Creator mobile app.
NFR-04	Availability	The application shall be available during clinic hours through the Zoho cloud.
NFR-05	Maintainability	Forms, reports and workflows shall be modifiable by a trained administrator without code deployment.
NFR-06	Auditability	Record changes shall be traceable through the Zoho Creator audit trail.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	Zoho Calendar (through a Zoho Creator connection)
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases of the three roles. The Receptionist registers patients, books appointments and prepares bills; the Doctor records treatments and updates appointment status; the Administrator maintains doctors, manages the application and monitors the dashboard. Table 3.5 and Table 3.6 describe two key use cases.

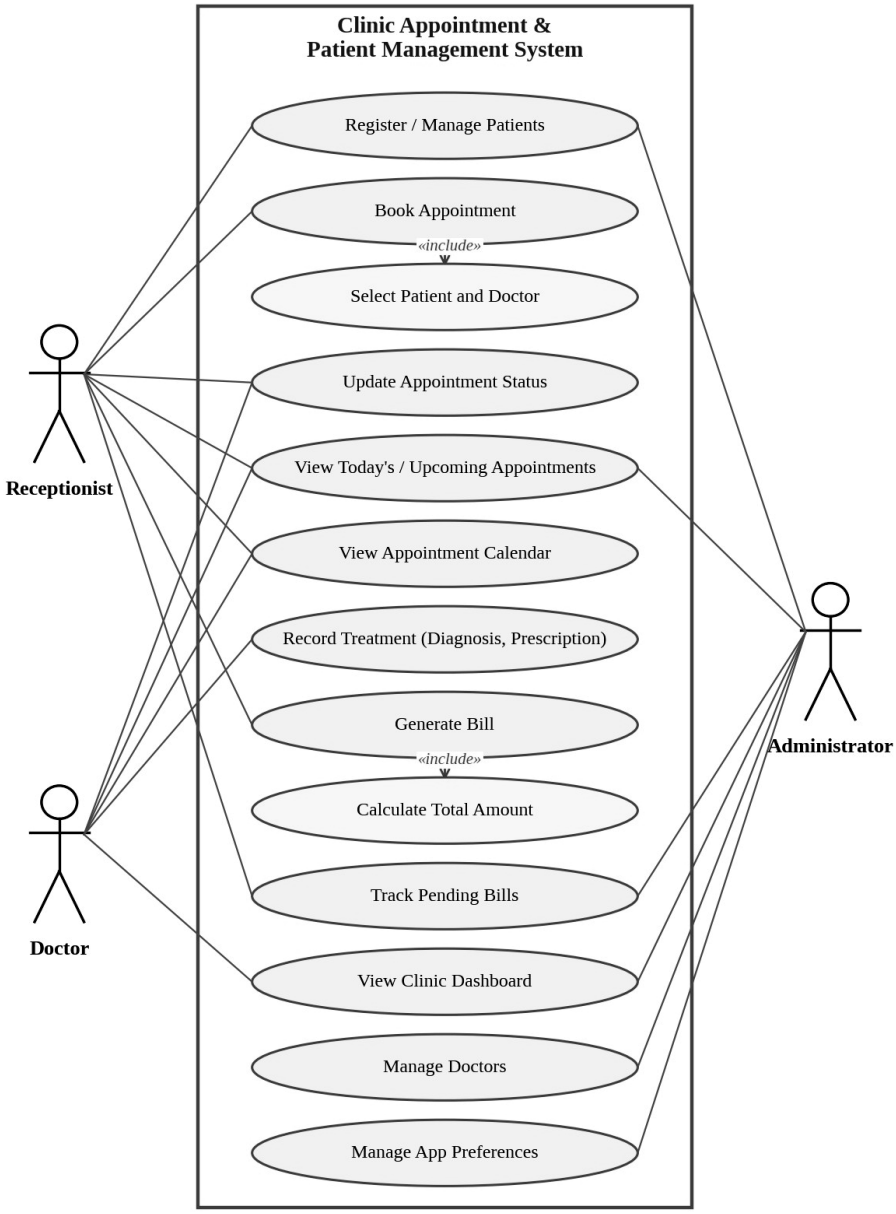


Figure 3.1: Use Case Diagram of the Clinic Appointment and Patient Management System

Table 3.5: Use Case Description – Book Appointment (UC-03)

Item	Description
Use Case ID	UC-03
Use Case Name	Book Appointment
Actor(s)	Receptionist
Pre-condition	The patient and the doctor already exist in the Patients and Doctors forms.
Main Flow	1. Receptionist opens the Appointments form. 2. Selects the patient and the doctor. 3. Enters date, time and reason for visit. 4. Submits the form. 5. System stores the record with status Scheduled and creates a Zoho Calendar event.
Alternate Flow	If a mandatory field is left empty, the form shows an error and the receptionist completes the field before submitting.

Item	Description
Post-condition	The appointment appears in the Upcoming Appointments report.
Related FR	FR-03, FR-04, FR-06, FR-10

Table 3.6: Use Case Description – Record Treatment and Bill (UC-05)

Item	Description
Use Case ID	UC-05
Use Case Name	Record Treatment and Bill
Actor(s)	Doctor, Receptionist
Pre-condition	An appointment exists for the visit.
Main Flow	1. Doctor opens the Treatments form and selects the appointment. 2. Enters diagnosis, prescription and notes. 3. Receptionist opens the Billing form, selects the patient and appointment. 4. Enters consultation fee and medicine charges; total amount is recorded. 5. Payment status is updated when paid.
Alternate Flow	If payment is not received, the status remains Pending.
Post-condition	The visit history of the patient contains the treatment and the bill.
Related FR	FR-07, FR-08, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 shows the architecture. Users of the three roles access the application through a web browser or the Zoho Creator mobile app. The application layer contains the forms, reports, the Clinic Dashboard page and the workflow with its Deluge script. All data is stored in the Zoho Creator cloud database, which is generated automatically from the forms. When an appointment is saved, the Deluge script uses the connection “zoho_calendar_connection” to create an event in Zoho Calendar; other Zoho services or REST APIs can be connected in the same way later.

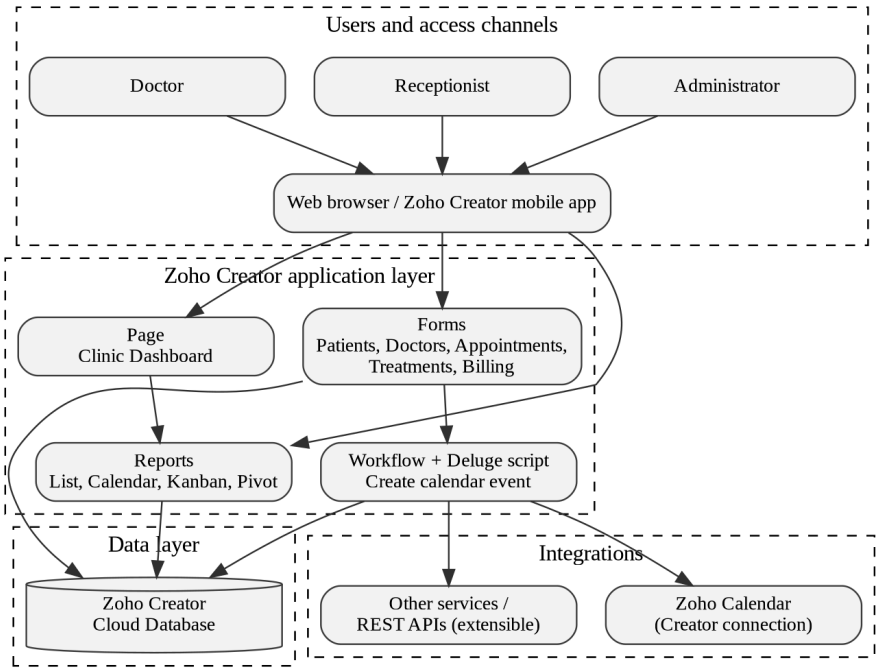


Figure 4.1: System Architecture of the Clinic Appointment and Patient Management System

4.2 Data Model

Figure 4.2 shows the entities and Table 4.1 summarises the relationships. A patient can have many appointments and many bills, and a doctor can have many appointments. A treatment and a bill belong to one appointment, while an appointment may have several treatment entries. In Zoho Creator these one-to-many relationships are implemented with lookup fields placed on the “many” side.

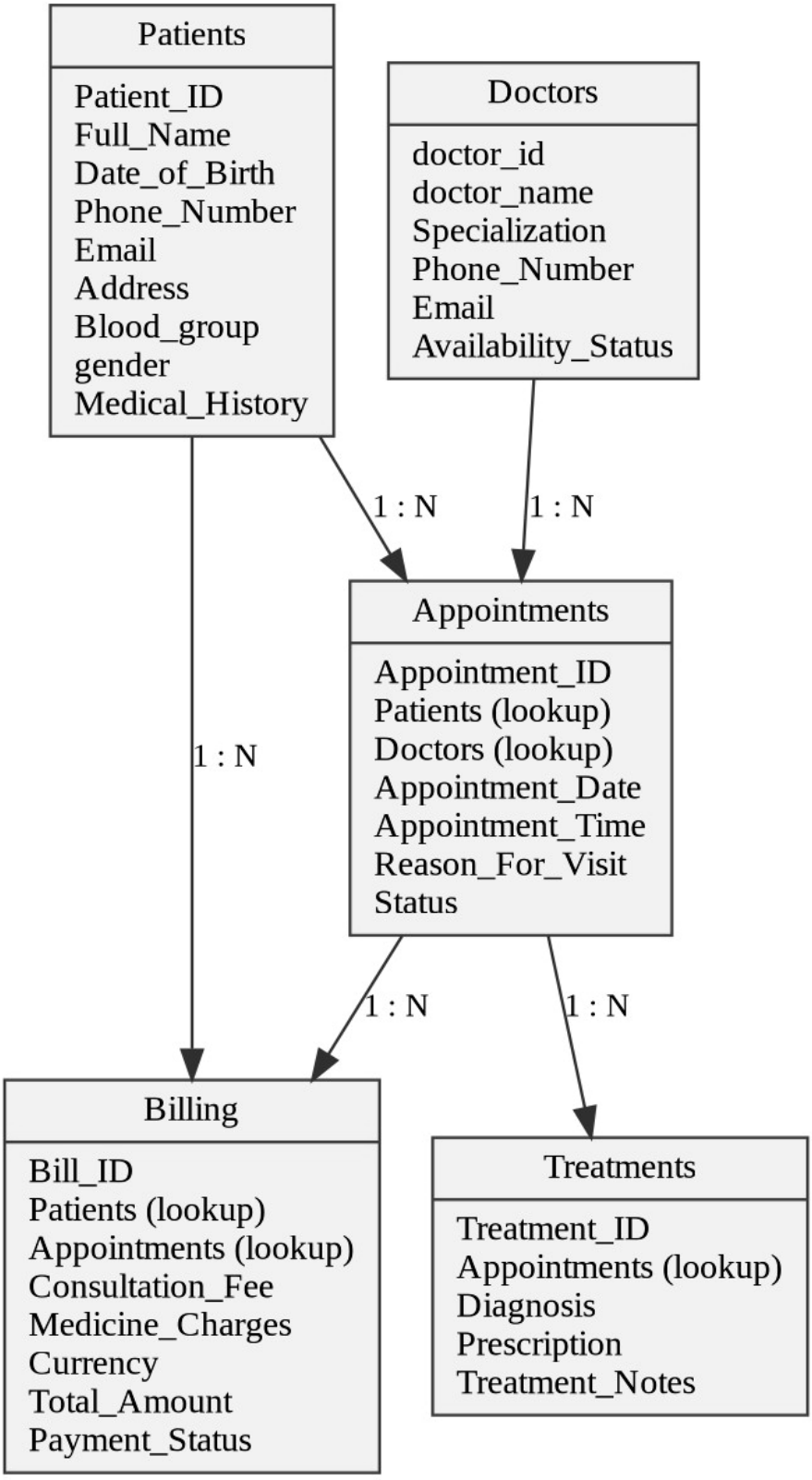


Figure 4.2: Entity–Relationship Diagram of the Application

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship	Implemented Using
Patients	Patient_ID, Full_Name, Phone_Number	Appointments	One-to-Many	Lookup field “Patients” in Appointments
Doctors	doctor_id, doctor_name, Specialization	Appointments	One-to-Many	Lookup field “Doctors” in Appointments
Appointments	Appointment_ID, Appointment_Date, Status	Treatments	One-to-Many	Lookup field “Appointments” in Treatments
Appointments	Appointment_ID, Appointment_Date, Status	Billing	One-to-Many	Lookup field “Appointments” in Billing
Patients	Patient_ID, Full_Name	Billing	One-to-Many	Lookup field “Patients” in Billing

4.3 Form Design

Each form has an Auto Number identifier and the fields listed in Table 4.2 to Table 4.6. Field labels, link names and types are as configured in the Zoho Creator form builder; the link name is the name by which a field is referred to in scripts and reports. Dropdown choices and mandatory settings are part of the design.

Table 4.2: Field Specification – Patients Form

Field Label	Link Name	Field Type	Mandatory	Validation / Choices	Remarks
Patient ID	Patient_ID	Auto Number	—	Start index 1	System generated
Full Name	Full_Name	Name	Yes	—	Patient’s name
Date of Birth	Date_of_Birth	Date	Yes	—	Used to derive age
Phone Number	Phone_Number	Phone	Yes	—	Contact number
Email	Email	Email	No	Valid e-mail format	Patient contact
Address	Address	Address	No	—	Residential address
Blood group	Blood_group	Drop Down	No	A+, A−, B+, B−, AB+, AB−, O+, O−	—
gender	gender	Drop Down	Yes	Male, Female, Other	—
Medical_History	Medical_History	Multi Line	No	—	Allergies, chronic illness

Table 4.3: Field Specification – Doctors Form

Field Label	Link Name	Field Type	Mandatory	Validation / Choices	Remarks
doctor_id	doctor_id	Auto Number	—	Start index 1	System generated

Field Label	Link Name	Field Type	Mandatory	Validation / Choices	Remarks
doctor name	doctor_name	Single Line	Yes	—	Doctor’s name
Specialization	Specialization	Drop Down	Yes	General Medicine, Pediatrics, Cardiology, Dermatology, Orthopedics, Other	—
Phone Number	Phone_Number	Phone	Yes	—	Contact number
Email	Email	Email	Yes	Valid e-mail format	—
Availability_Status	Availability_Status	Drop Down	Yes	Available, On Leave	Used when booking

Table 4.4: Field Specification – Appointments Form

Field Label	Link Name	Field Type	Mandatory	Validation / Choices	Remarks
Appointment_ID	Appointment_ID	Auto Number	—	Start index 1	System generated
Patients	Patients	Lookup	Yes	Source: Patients form	Links to patient
Doctors	Doctors	Lookup	Yes	Source: Doctors form	Links to doctor
Appointment_Date	Appointment_Date	Date	Yes	—	Date of visit
Appointment_Time	Appointment_Time	Time	Yes	—	Time of visit
Reason_For_Visit	Reason_For_Visit	Multi Line	Yes	—	Complaint / purpose
Status	Status	Drop Down	Yes	Scheduled, Completed, Cancelled; default Scheduled	Lifecycle of visit

Table 4.5: Field Specification – Treatments Form

Field Label	Link Name	Field Type	Mandatory	Validation / Choices	Remarks
Treatment_ID	Treatment_ID	Auto Number	—	Start index 1	System generated
Appointments	Appointments	Lookup	Yes	Source: Appointments form	Links to visit
Diagnosis	Diagnosis	Multi Line	Yes	—	Doctor’s diagnosis
Prescription	Prescription	Multi Line	No	—	Medicines advised
Treatment_Notes	Treatment_Notes	Multi Line	No	—	Additional notes

Table 4.6: Field Specification – Billing Form

Field Label	Link Name	Field Type	Mandatory	Validation / Choices	Remarks
Bill_ID	Bill_ID	Auto Number	—	Start index 1	System generated
Patients	Patients	Lookup	Yes	Source: Patients form	Links to patient
Appointments	Appointments	Lookup	Yes	Source: Appointments form	Links to visit
Consultation_Fee	Consultation_Fee	Currency	Yes	—	Doctor's fee
Medicine_Charges	Medicine_Charges	Currency	No	—	Cost of medicines
Currency	Currency	Currency	No	—	Additional charge field (default label)
Total_Amount	Total_Amount	Currency	Yes	Sum of the charges	Amount payable
Payment_Status	Payment_Status	Drop Down	Yes	Pending, Paid; default Pending	Payment tracking

4.4 Process and Workflow Design

Figure 4.3 shows the process for booking an appointment. The receptionist selects the patient and the doctor, enters the appointment date, time and reason for visit, and submits the form with the status Scheduled. A Deluge script then combines the date and time into a start time, adds 60 minutes to obtain the end time, and calls the Zoho Calendar connection to create an event titled "Patient Appointment" in the Asia/Kolkata time zone, so the visit also appears in the clinic's calendar.

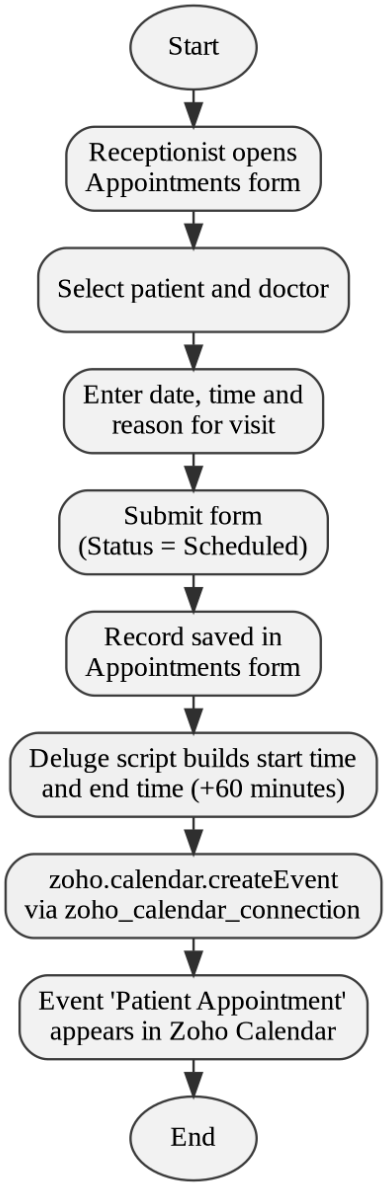


Figure 4.3: Workflow / Process Flow for Booking an Appointment

4.5 User Interface and Navigation Design

The application menu is organised by business area, as shown in Figure 4.4. Patients, Doctors, Treatments and Billings each contain the data-entry form and an “All …” report; Appointments additionally holds Today’s Appointments, Upcoming Appointments and an Appointment Calendar; Billings holds a Pending Bills report; App Preferences and the Clinic Dashboard sit at the top level. The Clinic Dashboard is a page that combines summary information about clinic activity.

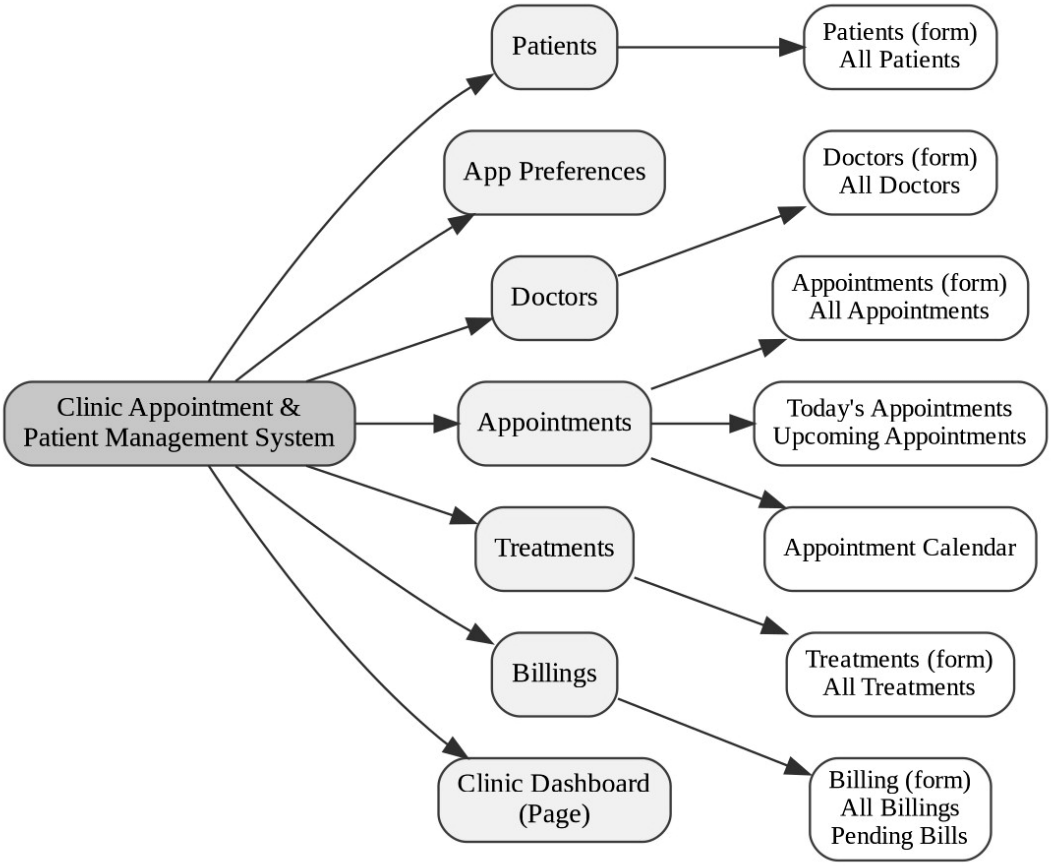


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Table 4.7 defines the permissions that will be implemented in Chapter 5 (A = Add, E = Edit, V = View, D = Delete, — = no access).

Table 4.7: Role–Permission Matrix

Component	Administrator	Receptionist	Doctor
Form: Patients	A E V D	A E V	V
Form: Doctors	A E V D	V	V
Form: Appointments	A E V D	A E V	V E (own)
Form: Treatments	V	V	A E V (own)
Form: Billing	A E V D	A E V	V
Report: Upcoming Appointments	V	V	V (own)
Page: Clinic Dashboard	V	—	V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Clinic Appointment & Patient Management System” was created in the Zoho Creator workspace under the application short code CAP, and its sections were organised to match the clinic’s work areas [7]. Table 5.1 lists the menu as it appears in the live application, and Figure 5.1 shows the application home. The application was built under a Zoho Creator trial, as shown by the “Trial expires in 15 days” banner at the top of every screen, and it can be previewed for desktop, tablet and mobile layouts using the device icons in the top bar.

Table 5.1: Application Menu Structure

Section	Items under the Section	Type of Items
Patients	Patients; All Patients	Form; report
App Preferences	App Preferences	Application settings
Doctors	Doctors; All Doctors	Form; report
Appointments	Appointments; All Appointments; All Appointments; Today’s Appointments; Upcoming Appointments; Appointment Calendar	Form; reports (including a calendar view)
Treatments	Treatments; All Treatments	Form; report
Billings	Billing; All Billings; Pending Bills	Form; reports
Clinic Dashboard	Clinic Dashboard	Page

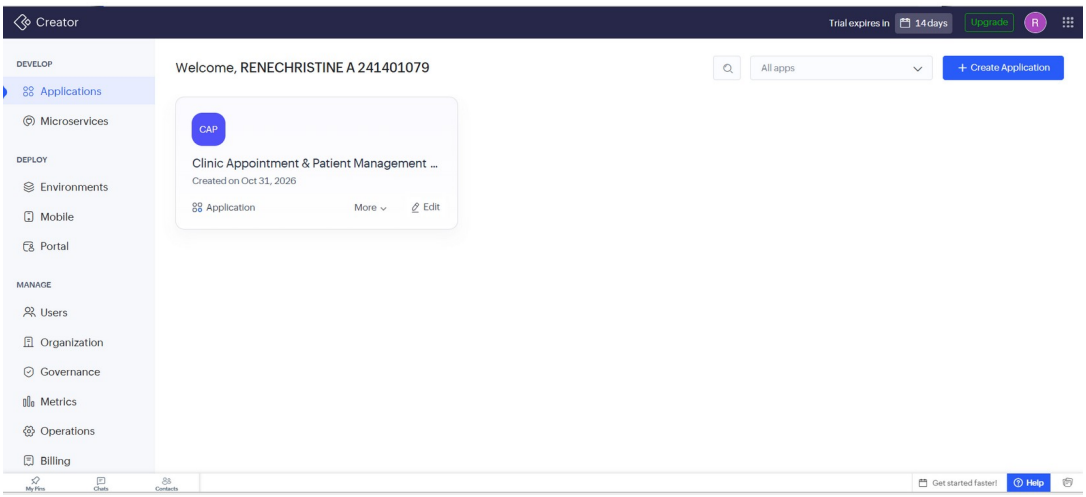


Figure 5.1: Application Home – Navigation Sections and Upcoming Appointments Report

5.2 Forms and Data Capture

Five forms hold the clinic’s data. Each form begins with an Auto Number field that generates a unique identifier for every record. Twelve distinct field types are used

across the forms — Auto Number, Name, Single Line, Multi Line, Date, Time, Phone, Email, Address, Drop Down, Lookup and Currency — and lookup fields create the relationships described in Section 4.2. Each form is described below with its form-builder view and then its live-mode view.

5.2.1 Patients Form

The Patients form stores the demographic and medical details of each patient. Patient ID is an Auto Number field with link name Patient_ID and start index 1. Full Name uses the Name field type, which captures the first name and the last name separately. Date of Birth uses the Date type and is displayed in the dd-MMM-yyyy format. Phone Number uses the Phone type with a country-code selector (+91 for India), and Email uses the Email type. The Address field is split into Address Line 1, Address Line 2, City / District, State / Province, Postal Code and Country. Blood group and gender are Drop Down fields, and Medical_History is a Multi Line field for allergies and chronic conditions. Figure 5.2 shows the form in the builder with Patient ID selected, and Figure 5.3 shows the live form with a sample patient, John Smith, entered.

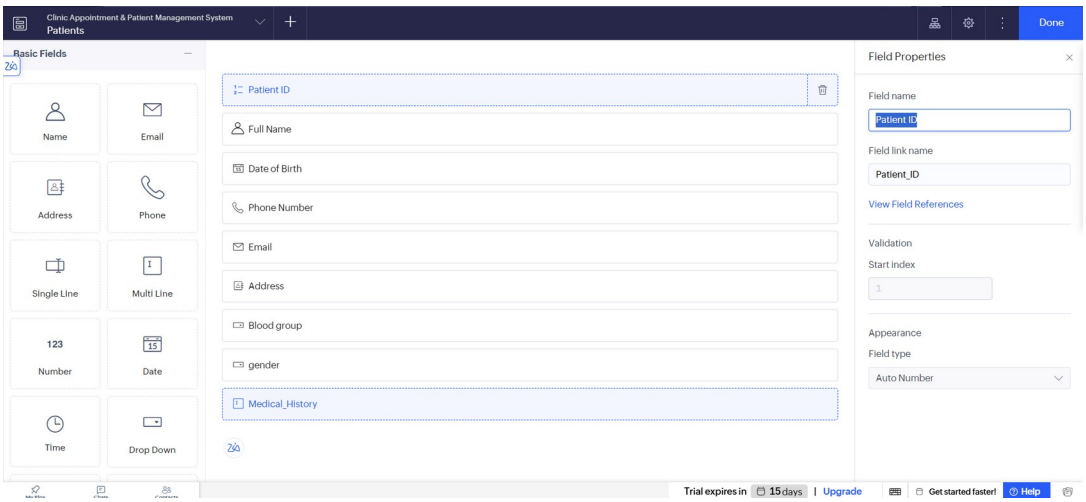


Figure 5.2: Patients Form in the Form Builder

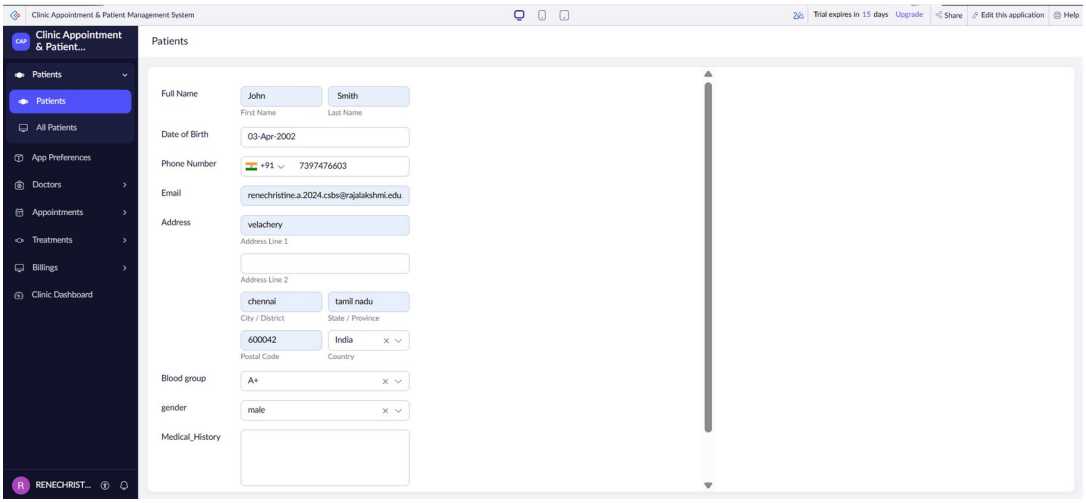


Figure 5.3: Patients Form in Live Mode

5.2.2 Doctors Form

The Doctors form maintains the list of consulting doctors. The doctor_id field is an Auto Number identifier; doctor name is a Single Line field; Specialization and Availability_Status are Drop Down fields; Phone Number and Email use the Phone and Email types. The Availability_Status field lets the front desk see whether a doctor can currently be booked. In the sample data the doctor “mary” is entered with the specialization Dentist, the availability Available and a +91 phone number. Figure 5.4 and Figure 5.5 show the builder and live views.

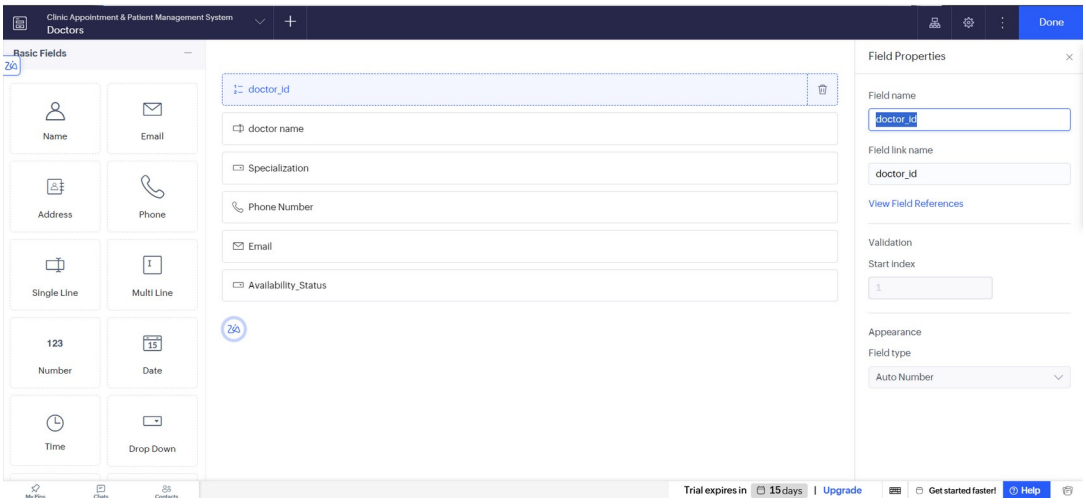


Figure 5.4: Doctors Form in the Form Builder

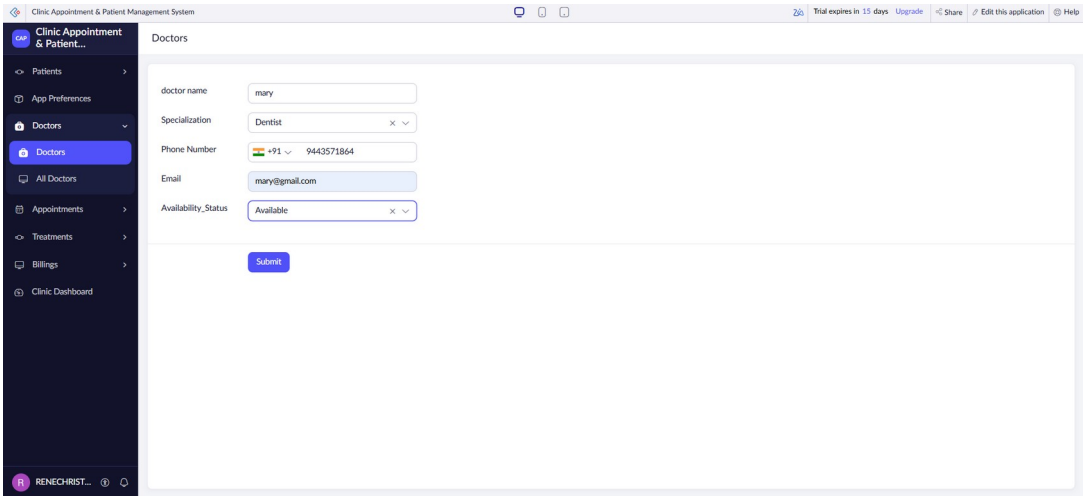


Figure 5.5: Doctors Form in Live Mode

5.2.3 Appointments Form

The Appointments form is the central transaction form. Appointment_ID is generated automatically. The Patients and Doctors fields are lookup fields shown as drop-down lists that pick records from the Patients and Doctors forms, so a patient or doctor is never retyped. Appointment_Date uses the Date type and Appointment_Time the Time type (stored as hours, minutes and seconds). Reason_For_Visit is a Multi Line field and Status is a Drop Down whose value is Scheduled for a newly booked visit. In the sample record, the patient John Smith is booked with the doctor “rene” on 02-Nov-2026 at 21:30:21 for a regular checkup. Figure 5.6 and Figure 5.7 show the form.

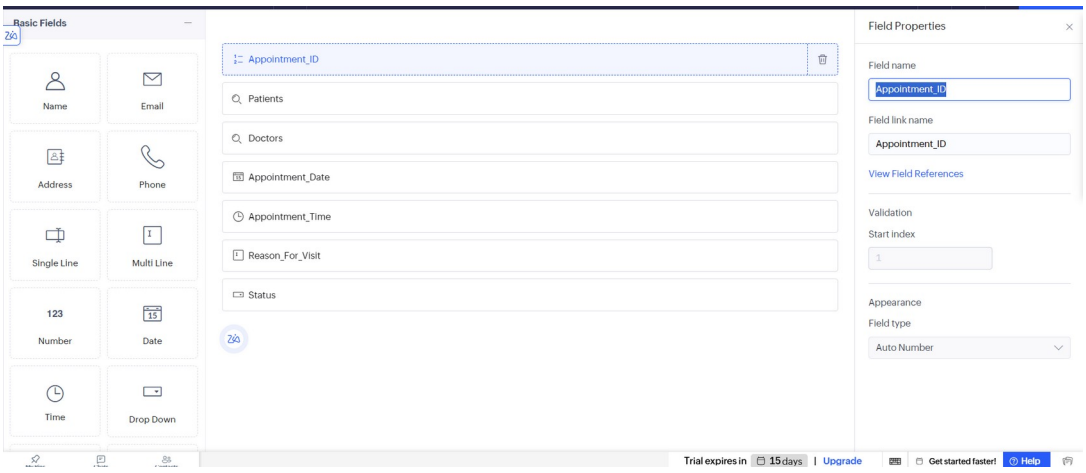


Figure 5.6: Appointments Form in the Form Builder

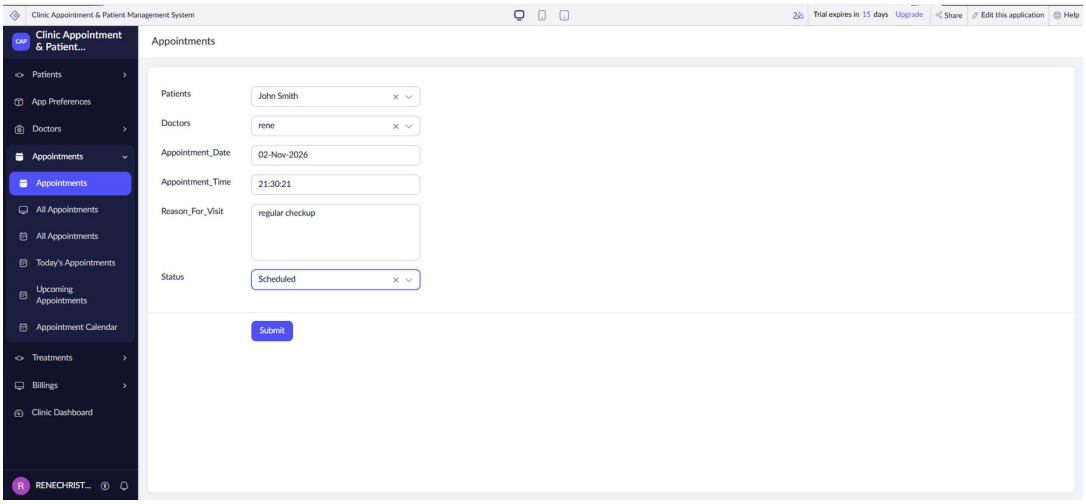


Figure 5.7: Appointments Form in Live Mode

5.2.4 Treatments Form

The Treatments form records the clinical outcome of a visit. Treatment_ID is an Auto Number field, and the Appointments lookup ties the treatment to the visit; in live mode this lookup is displayed as radio buttons that list the appointment IDs (1, 2 and 3 in the sample data), from which the patient and doctor can be traced. Diagnosis, Prescription and Treatment_Notes are Multi Line fields. In the sample record for appointment 1, the diagnosis is “more cavity”, the prescription is “root canal”, and the notes advise a regular check, brushing the teeth twice a day and avoiding hard foods. Figure 5.8 and Figure 5.9 show the form.

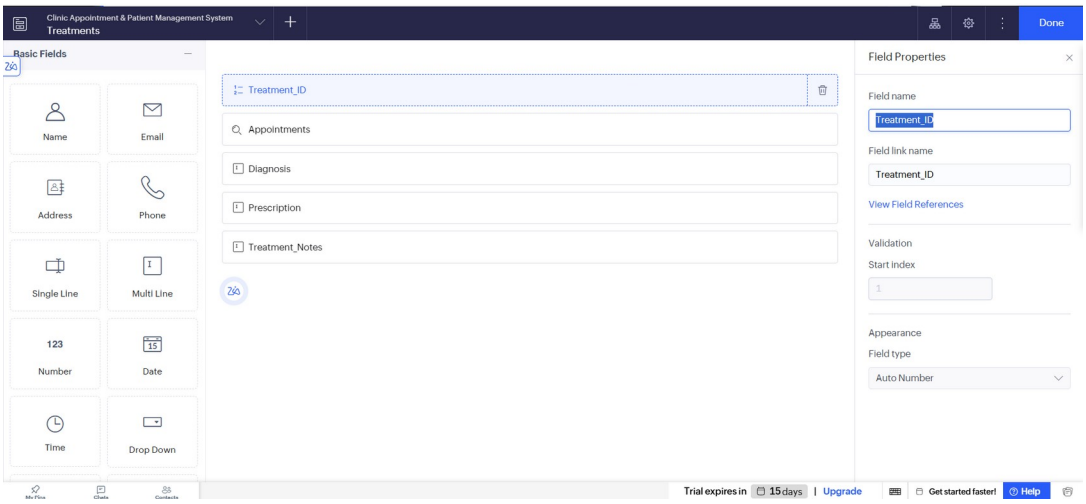


Figure 5.8: Treatments Form in the Form Builder

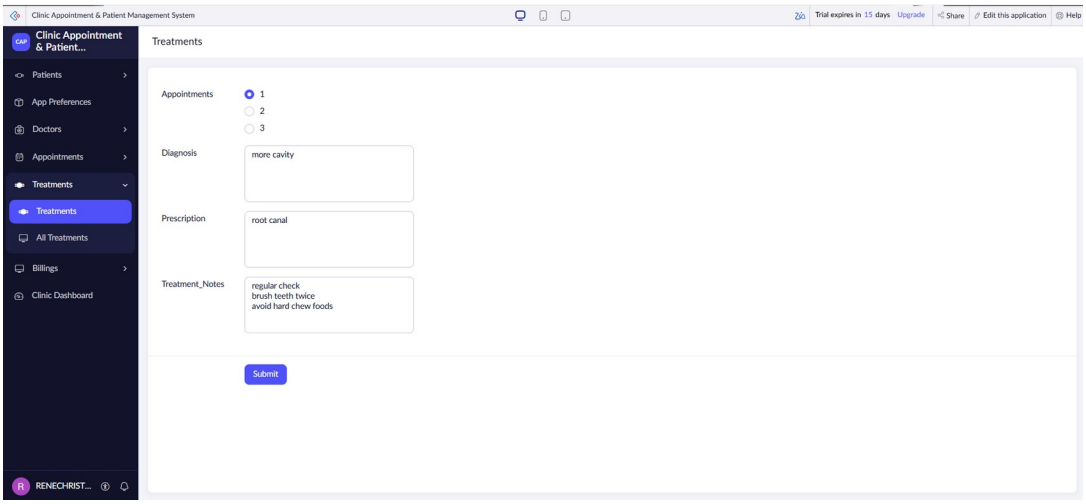


Figure 5.9: Treatments Form in Live Mode

5.2.5 Billing Form

The Billing form records the charges for a visit. Bill_ID is an Auto Number field; the Patients and Appointments lookups link the bill to the patient and the visit. Consultation_Fee, Medicine_Charges, Currency and Total_Amount use the Currency field type, and Payment_Status is a Drop Down used to track whether the bill is pending or paid. In the sample bill for John Smith and appointment 1, the consultation fee is 600 and the medicine charges are 500, giving a Total_Amount of 1,100.00 with the payment status Pending. Figure 5.10 and Figure 5.11 show the form.

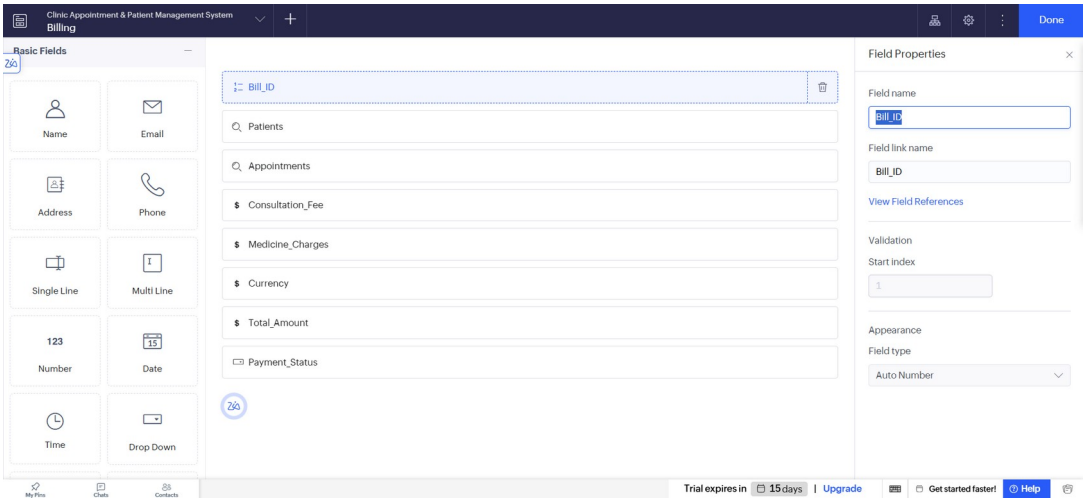


Figure 5.10: Billing Form in the Form Builder

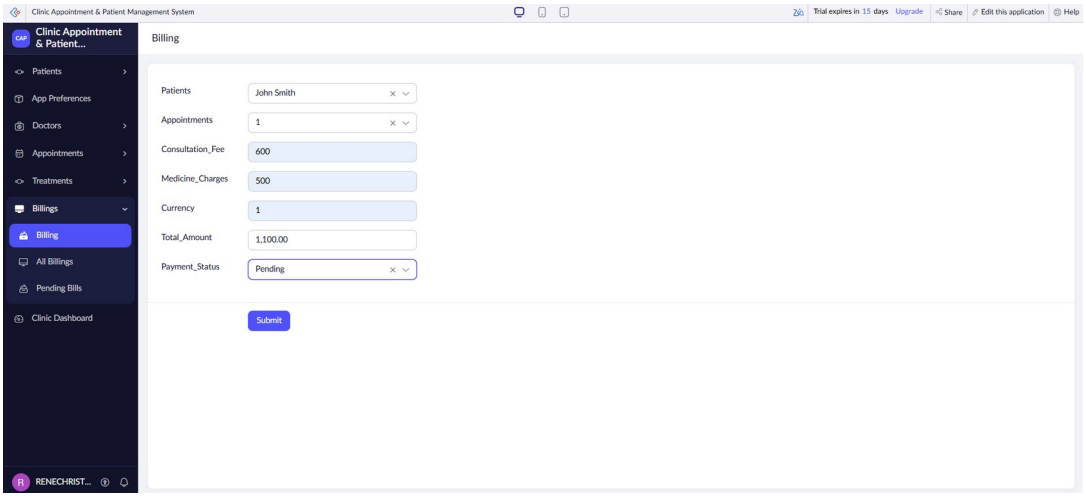


Figure 5.11: Billing Form in Live Mode

5.2.6 Sample Data Captured

Table 5.2 summarises the sample records entered through the five forms during development, which are used in the screenshots of this chapter.

Table 5.2: Sample Records Entered in the Forms

Form	Sample Record
Patients	Full Name: John Smith; Date of Birth: 03-Apr-2002; Phone: +91 (India); Address: Velachery, Chennai, Tamil Nadu, 600042, India; Blood group: A+; gender: male
Doctors	doctor name: mary; Specialization: Dentist; Phone: +91 (India); Availability_Status: Available
Appointments	Patients: John Smith; Doctors: rene; Appointment_Date: 02-Nov-2026; Appointment_Time: 21:30:21; Reason_For_Visit: regular checkup; Status: Scheduled
Treatments	Appointments: 1; Diagnosis: more cavity; Prescription: root canal; Treatment_Notes: regular check, brush teeth twice, avoid hard chew foods
Billing	Patients: John Smith; Appointments: 1; Consultation_Fee: 600; Medicine_Charges: 500; Total_Amount: 1,100.00; Payment_Status: Pending

5.3 Reports

Zoho Creator generated a list report for each form, and further reports were created on the Appointments and Billing forms to give each role a ready-made view of the data. Table 5.3 lists the reports shown in this chapter with their source form, filter and the roles for which they are intended (Table 4.7). Sorting, grouping and custom buttons were not configured for any report.

Table 5.3: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Roles
All Patients	List (auto-created)	Patients	None	Administrator; Receptionist; Doctor (view)
All Doctors	List (auto-created)	Doctors	None	Administrator; Receptionist; Doctor (view)
All Appointments	List	Appointments	None	Administrator; Receptionist; Doctor (own)
Today's Appointments	List	Appointments	Filter on Appointment_Date (today)	Administrator; Receptionist; Doctor (own)
Upcoming Appointments	List	Appointments	Filter on Appointment_Date (after today)	All roles; Doctor (own)
All Treatments	List (auto-created)	Treatments	None	Administrator; Receptionist (view); Doctor
Pending Bills	List	Billing	Filter: bills not yet paid	Administrator; Receptionist; Doctor (view)

5.3.1 Patient and Doctor Reports

All Patients lists the registered patients with their Full Name, Date of Birth, Phone Number, Email, Address and gender, and it is the report the receptionist opens to find an existing patient before booking an appointment (Figure 5.12). All Doctors lists the doctor_id, doctor name, Specialization, Phone Number, Email and Availability_Status of each doctor (Figure 5.13). The screenshot shows three doctors: kevin christo (Pediatrician), pradeep (General Physician) and rene (General Physician), all Available. Both are default list reports created with the forms, and clicking a row opens the full record.

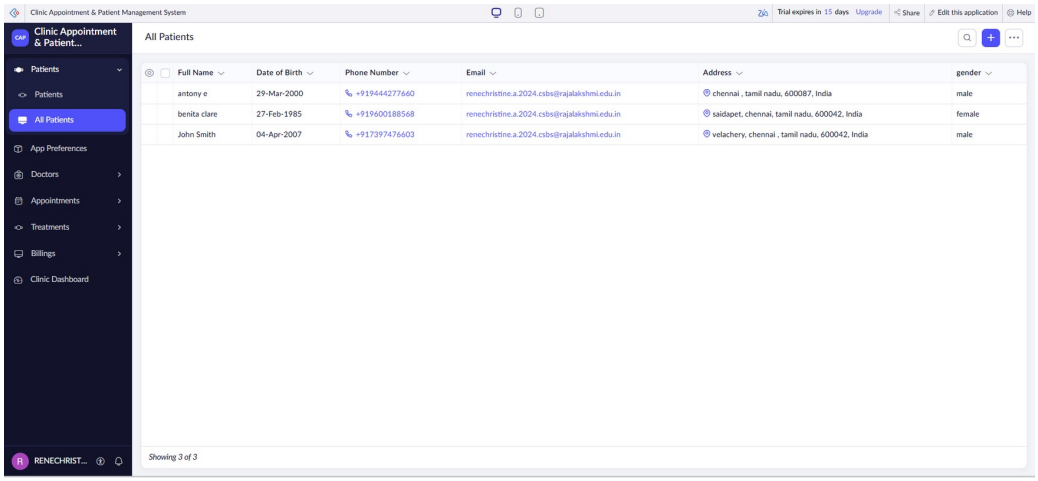


Figure 5.12: All Patients Report

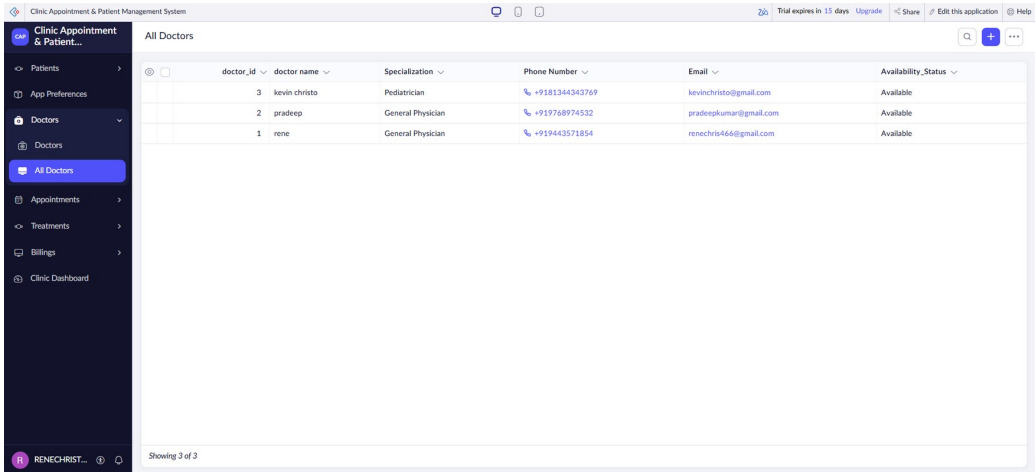


Figure 5.13: All Doctors Report

5.3.2 Appointment Reports

Three reports are built on the Appointments form. All Appointments shows every booking with Appointment_ID, Patients, Doctors, Appointment_Date, Appointment_Time, Reason_For_Visit and Status (Figure 5.15); the screenshot shows appointments 6, 5 and 4, all Scheduled. The date-based reports restrict the records using the Filter option of the report builder, whose Organize Records panel offers Filter, Grouping and Sorting and, under User Options, Quick Filters, Custom Filters and Print & PDF. A filter row is defined by choosing a field (Appointment_Date), an operator (equals) and a value, and rows can be combined with AND or OR (Figure 5.14).

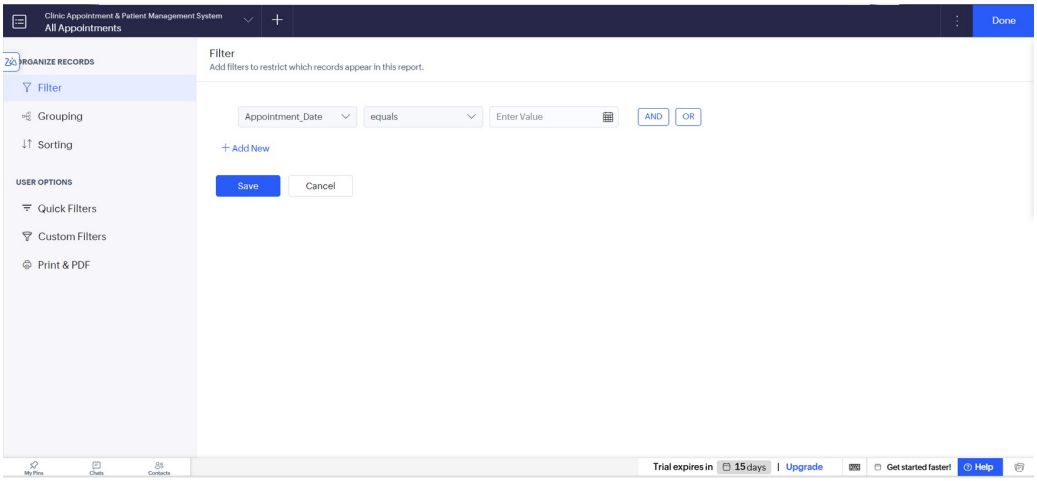


Figure 5.14: Filter Configuration of an Appointments Report

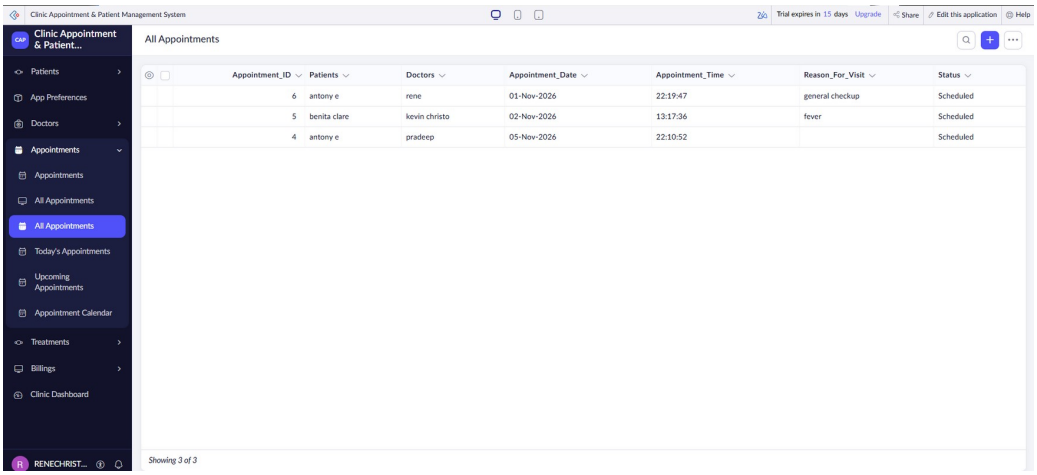


Figure 5.15: All Appointments Report

Today's Appointments is the front-desk view of the current day. In the screenshot it contains appointment 7 for the patient Sarah Smith with the doctor pradeep on 31-Oct-2026 at 22:32:34 for stomach pain, and the message "Appointment Added Successfully" confirms that the record had just been saved (Figure 5.16). Upcoming Appointments lists the visits that fall after the current date, so appointment 7 is not shown and the report lists appointments 6, 5 and 4 dated 01-Nov-2026, 02-Nov-2026 and 05-Nov-2026 (Figure 5.17). This report is also embedded in the Clinic Dashboard (Section 5.4).

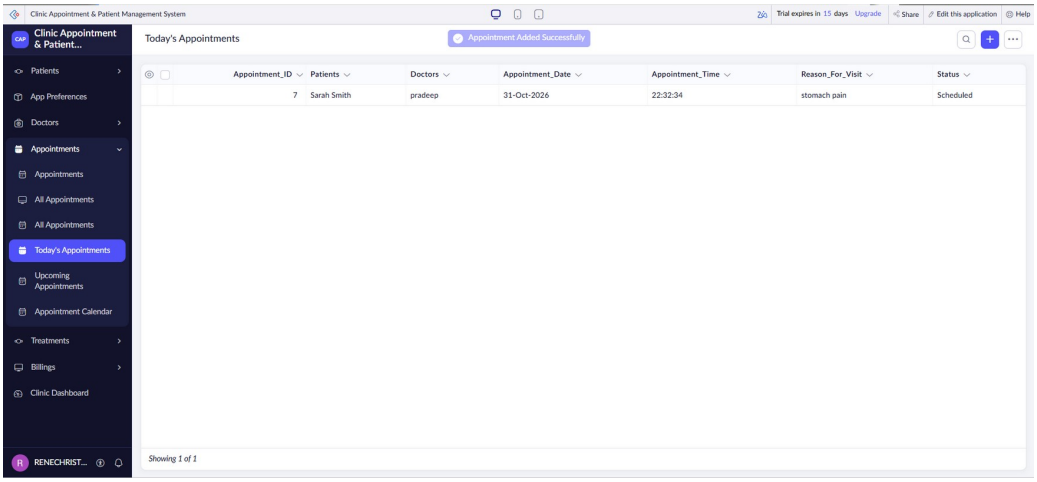


Figure 5.16: Today's Appointments Report

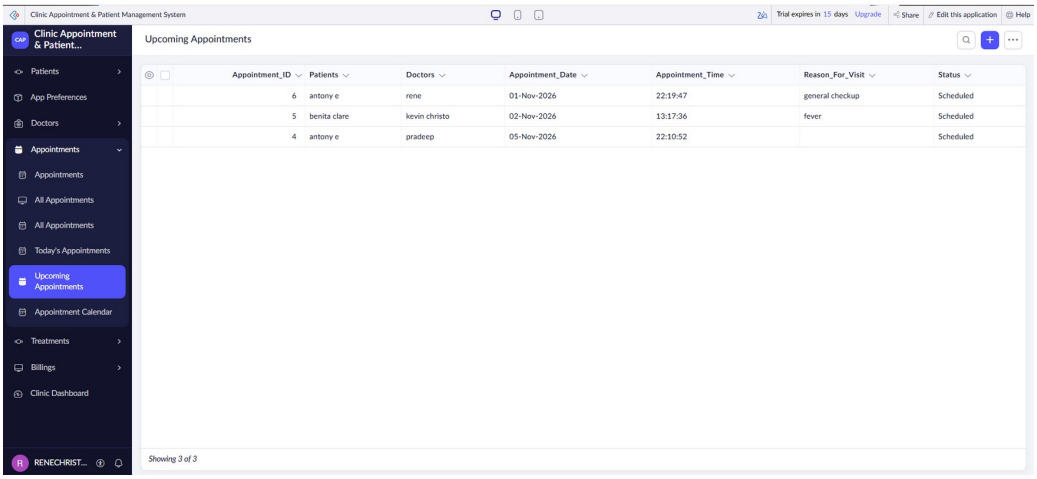


Figure 5.17: Upcoming Appointments Report

5.3.3 Treatment Report

Treatments are entered through the Treatments form, where the Appointments lookup is displayed as radio buttons listing the appointment IDs 4, 5, 6 and 7 (Figure 5.18). All Treatments lists Treatment_ID, Appointments, Diagnosis, Prescription and Treatment_Notes for every saved treatment, with a horizontal scroll bar because the text columns are wide (Figure 5.19). The screenshot shows four records, one for each of appointments 4 to 7: viral fever (appointment 4), Type 2 Diabetes Mellitus (appointment 5), common cold with cough and nasal congestion (appointment 6) and gastritis (appointment 7). Each record carries a prescription entry and advice notes, so a doctor can read a patient's earlier diagnosis and advice from one report.

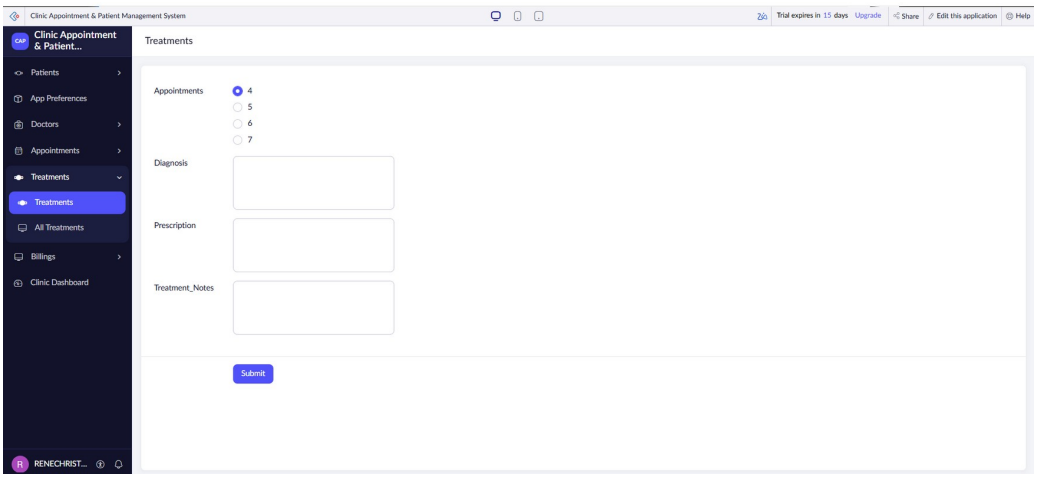


Figure 5.18: Treatments Form in Live Mode with Appointment Selection

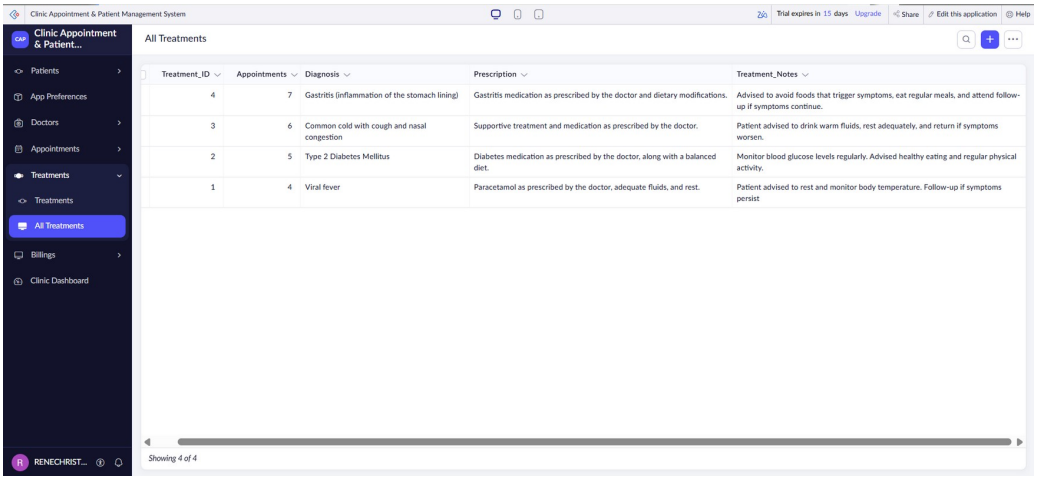


Figure 5.19: All Treatments Report

5.3.4 Billing Report

Pending Bills lists the bills that are still to be collected, with Bill_ID, Patients, Appointments, Consultation_Fee, Medicine_Charges, Currency and Total_Amount. Its first screenshot shows three bills: bill 4 for Sarah Smith (appointment 7) with a total of ₹1,200.00, bill 2 for antony e (appointment 4) with ₹1,400.00 and bill 1 with ₹1,600.00 (Figure 5.20). After a further bill was entered, bill 5 for John Smith (appointment 5, ₹1,600.00) appeared at the top and the footer changed to “Showing 4 of 4” (Figure 5.21), which shows that the report refreshes as new pending bills are saved. In every row the Total_Amount equals the sum of Consultation_Fee and Medicine_Charges, and the four pending bills add up to ₹5,800.00.

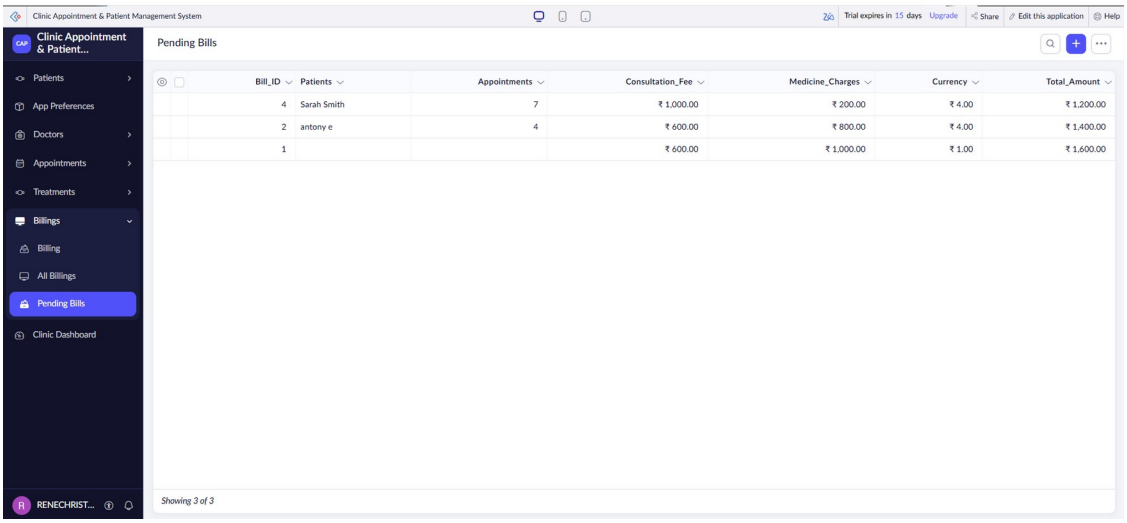


Figure 5.20: Pending Bills Report (3 of 3 Records)

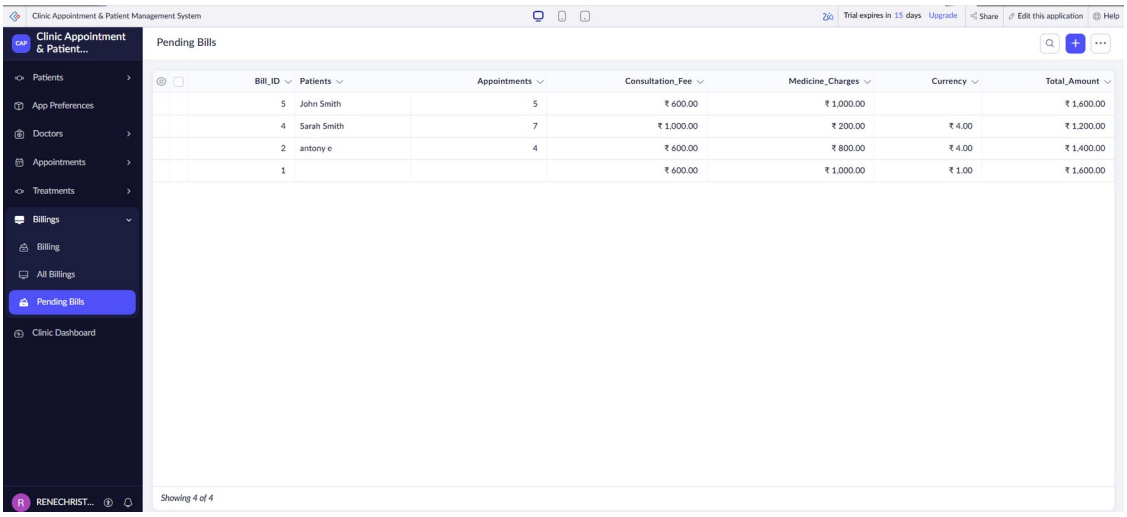


Figure 5.21: Pending Bills Report after a New Bill Was Added (4 of 4 Records)

5.4 Pages and Dashboards

The Clinic Dashboard is a page built in the UI Builder, which opens with an empty canvas marked “Drag elements here” and an element palette offering Panel, Board, Chart, Gauge, Search, Form, Report, Snippets, Button, Widgets and Viewer (Figure 5.22). It gives the Administrator and the Doctor a single screen that summarises the clinic’s activity (Table 4.7); the Receptionist has no access to the page. The dashboard contains four summary cards, two charts and two embedded reports, listed in Table 5.4.

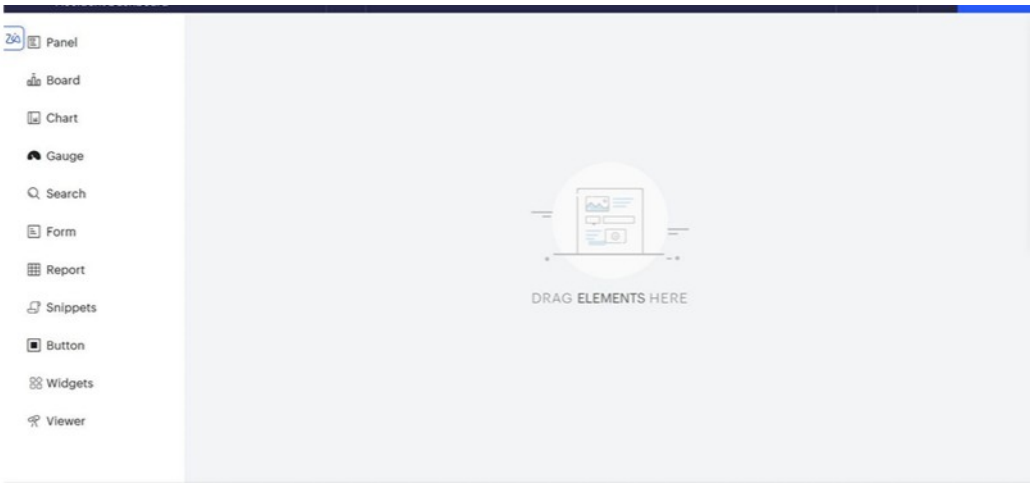


Figure 5.22: UI Builder Element Palette of the Clinic Dashboard Page

Table 5.4: Components of the Clinic Dashboard

Component	Type	Data Shown	Value in the Screenshot
Total Patients	Summary card	Number of records in the Patients form (“Registered patients”)	5
Total Doctors	Summary card	Number of records in the Doctors form (“Available doctors”)	3
Total Appointments	Summary card	Number of records in the Appointments form (“All appointments”)	4
Pending Bills	Summary card	Number of pending bills (“Outstanding bills”)	4
Appointments by Status	Chart (pie)	Appointments grouped by Status	One slice: Scheduled
Appointments by Date	Chart (bar)	Count of Appointment_ID by month of Appointment_Date	Oct 2026: 1; Nov 2026: 3
Upcoming Appointments	Embedded report	Upcoming Appointments report	Appointments 6, 5 and 4 (3 of 3)
Pending Bills	Embedded report	Pending Bills report	Bills 5, 4, 2 and 1 (4 of 4)

The summary cards at the top give the four headline figures of the clinic: 5 registered patients, 3 doctors, 4 appointments and 4 outstanding bills (Figure 5.23). Below them, the Appointments by Status pie chart shows a single slice labelled Scheduled because every appointment in the sample data still has that status, and the Appointments by Date bar chart shows one appointment in October 2026 and three in November 2026, which agree with the appointment records of Section 5.3.2. Scrolling down, the Upcoming Appointments and Pending Bills reports are embedded side by side, each with its own search box, add button and horizontal scroll bar, so the front desk can open a record or add a new one without leaving the dashboard (Figure 5.24). The dashboard figures agree with the underlying reports: the four pending bills are the four rows of Figure 5.21, and the three upcoming appointments are those of Figure 5.17.

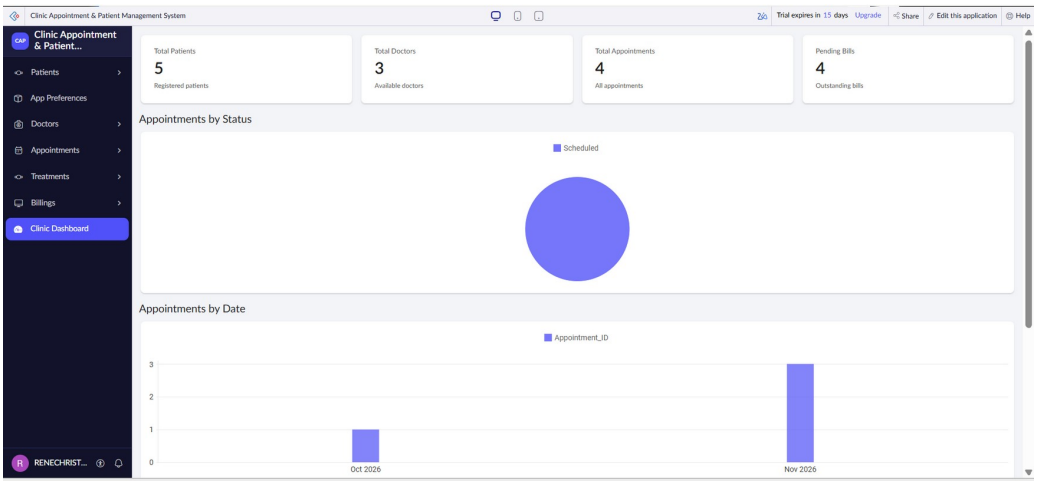


Figure 5.23: Clinic Dashboard – Summary Cards and Charts

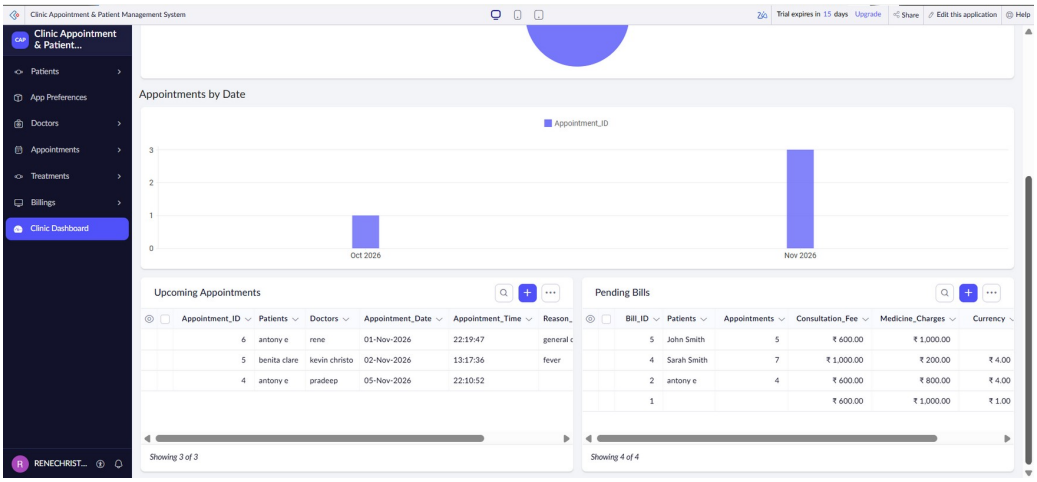


Figure 5.24: Clinic Dashboard – Upcoming Appointments and Pending Bills Reports

5.5 Deluge Scripting

Deluge scripts are attached as actions to the form workflows of Section 5.6. The script that integrates the application with Zoho Calendar is written in the Deluge editor of the "Create Calendar Event" workflow on the Appointments form; the editor, with its list of script blocks (Condition, Data Access, Miscellaneous) on the left, is shown in Figure 5.25. Table 5.5 lists the script and Code 5.1 shows its code; the complete code is also given in Appendix C.

Table 5.5: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Appointments → Created → Successful form submission (workflow Create Calendar Event)	Create a 60-minute Zoho Calendar event for each appointment	Map, put, toString, toDateTime, addMinutes, zoho.calendar.createEvent, info

Code 5.1: Calendar event script (DS-01)

```
// DS-01 | Appointments | Successful form submission | Create Calendar
Event
eventDetails = Map();
eventDetails.put("estatus","added");
eventDetails.put("title","Patient Appointment");
dateText = input.Appointment_Date.toString("dd-MMM-yyyy");
timeText = input.Appointment_Time.toString();
appointmentDateTime = (dateText + " " + timeText).toDateTime();
endDateTime = appointmentDateTime.addMinutes(60);
dateTime = Map();
dateTime.put("timezone","Asia/Kolkata");
dateTime.put("start",appointmentDateTime.toString("yyyyMMdd'T'HHmmss")
+ "+0530");
dateTime.put("end",endDateTime.toString("yyyyMMdd'T'HHmmss") +
"+0530");
eventDetails.put("dateandtime",dateTime);
event = Map();
event.put("eventdata",eventDetails);
response =
zoho.calendar.createEvent("a5dea53efcc54b709ba293ac40d9bf12",event,"zo
ho_calendar_connection");
info appointmentDateTime;
info response;
```

The script first builds a map of event details with the title "Patient Appointment" and the status "added". It then converts the Appointment_Date value to the text form dd-MMM-yyyy, joins it with the Appointment_Time value and converts the result into a date-time, which becomes the start time of the event. The end time is obtained by adding 60 minutes to the start time. Both times are written in the yyyyMMdd'T'HHmmss format with the +0530 offset of India, and they are placed in a map together with the time zone Asia/Kolkata. Finally the script wraps the event details in a map named eventdata and

calls zoho.calendar.createEvent with the identifier of the clinic's calendar and the connection zoho_calendar_connection (Section 5.7). The two info statements write the computed start time and the response of the calendar service to the execution log, which is how the call is checked during debugging. The workflow is Enabled and the script was saved without syntax errors, so it runs every time a new appointment is submitted.

The other workflows of Section 5.6 (confirmation e-mail, double-booking check, total calculation and reminder schedule) each contain one action configured in the workflow builder. Standalone Deluge functions, loops and REST calls with invokeurl were not needed for this application.

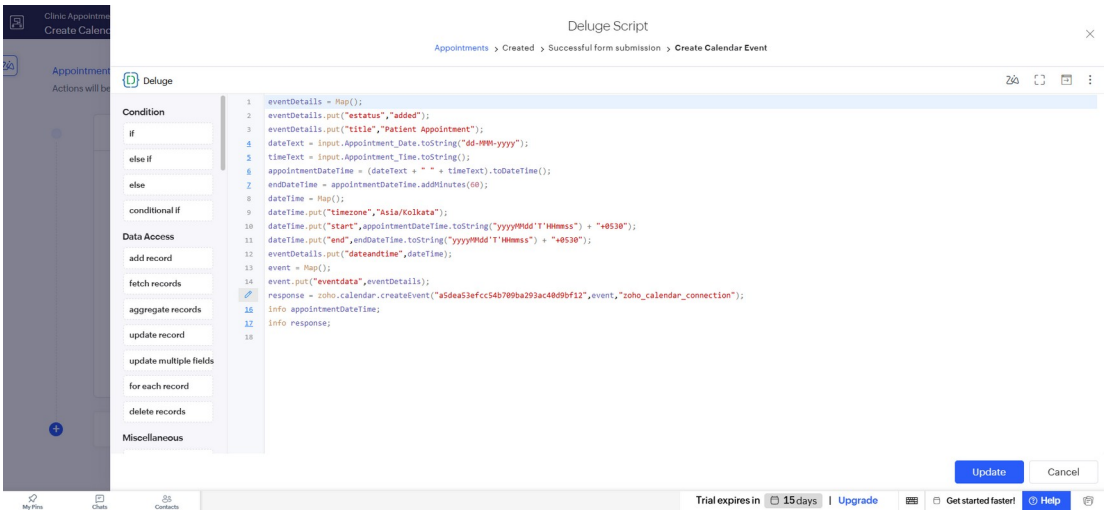


Figure 5.25: Deluge Editor with the Calendar Event Script (DS-01)

5.6 Workflows and Automation

Form workflows and one schedule were used. The Workflow tab of the builder also offers Approvals, Payments, Blueprints, Batch workflows and Functions; these were not needed for the delivered scope. The Form workflows tab lists the forms that have workflows (Appointments and Billing) and the forms without workflows (Doctors, Patients and Treatments), and each workflow is shown with its event, the number of actions, its status and the date of creation. All workflows were created on 31-Oct-2026 and are Enabled.

On the Appointments form, two workflows run on the event Created → Successful form submission, so they act after a new appointment has been saved: Appointment Confirmation Email and Create Calendar Event. A third workflow, Prevent Double Booking, runs on the event Created or Edited → Validations on form submission, so it checks the appointment before it is saved and can stop a booking that should not be accepted (Figure 5.26). On the Billing form the workflow Auto Calculate Total Amount runs on Created or Edited → User input of Medicine_Charges, so the total amount is worked out as soon as the medicine charges are entered (Figure 5.27). In the sample data

each total equals the consultation fee plus the medicine charges (for example ₹600.00 + ₹1,000.00 = ₹1,600.00).

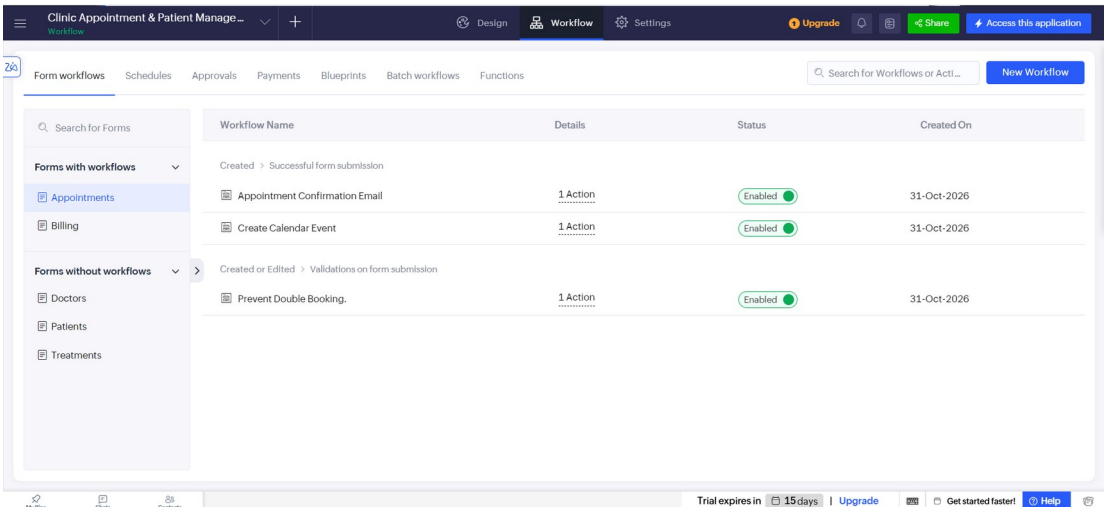


Figure 5.26: Form Workflows of the Appointments Form

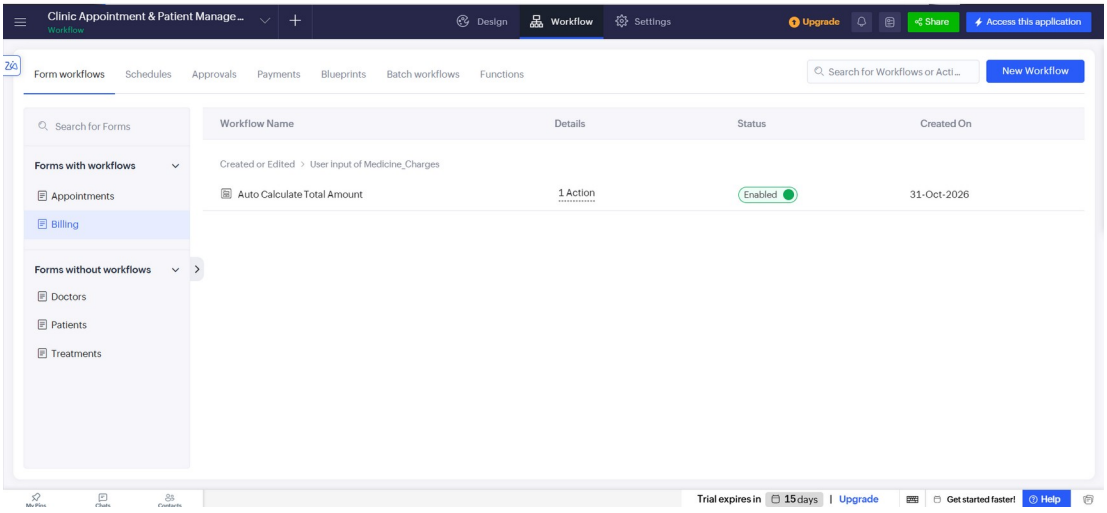


Figure 5.27: Form Workflow of the Billing Form

The Schedules tab holds one custom schedule, Appointment Reminder Scheduler, which has one action, starts on 31-Oct-2026 at 08:00:00 and is Enabled (Figure 5.28). A schedule runs without any user action at the set time, so it is used for the reminder that would otherwise have to be sent by phone from the front desk. The schedule is set to run at 08:00:00, the start of the clinic day.

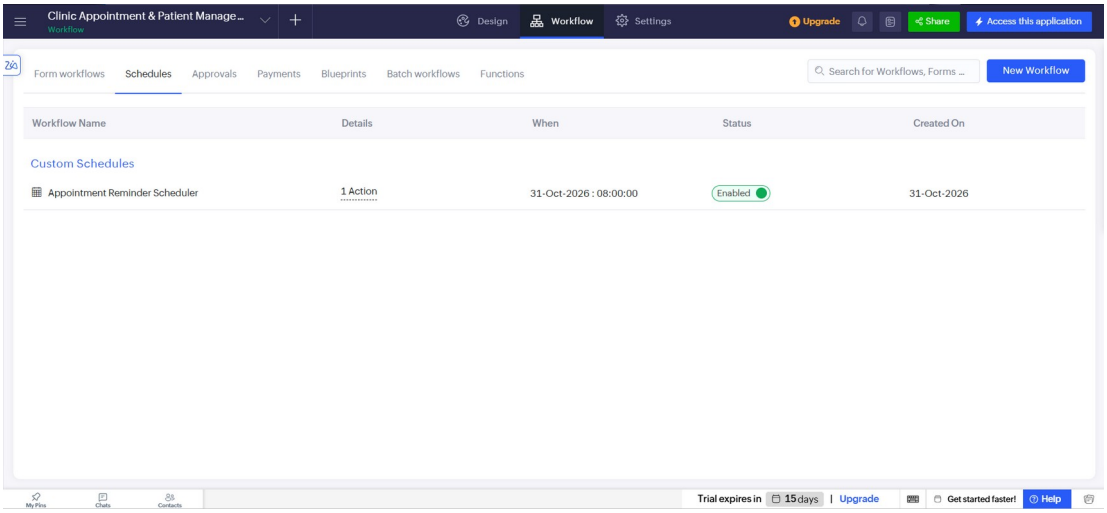


Figure 5.28: Custom Schedule Appointment Reminder Scheduler

Table 5.6: Workflows Implemented

Workflow Name	Type	Trigger / Event	Action(s)	Script Ref.
Appointment Confirmation Email	Form workflow	Appointments → Created → Successful form submission	1 action: confirmation e-mail for the booked appointment	—
Create Calendar Event	Form workflow	Appointments → Created → Successful form submission	1 action: Deluge script creating a 60-minute calendar event	DS-01
Prevent Double Booking	Form workflow	Appointments → Created or Edited → Validations on form submission	1 action: validation that checks the appointment before it is saved	—
Auto Calculate Total Amount	Form workflow	Billing → Created or Edited → User input of Medicine_Charges	1 action: sets Total_Amount from the fee and charges entered	—
Appointment Reminder Scheduler	Schedule	Custom schedule starting 31-Oct-2026, 08:00:00	1 action: appointment reminder	—

5.7 Integrations

Zoho Calendar. A connection named “Zoho Calendar Connection” with the link name zoho_calendar_connection exists under Microservices → Connections and shows the status Connected, while the system entry “Zoho Calendar” in the same list shows Not Connected (Figure 5.29). The connection details page shows the service name Zoho Calendar, the permissions “Create Calendar, Create Event, Get all Calendar”, the authorization “Same for all the environments”, the authorization owner Administrator and an authorized Zoho Calendar account. Connection access is switched On for the

Clinic Appointment & Patient Management System, which allows the application to use the connection (Figure 5.30).

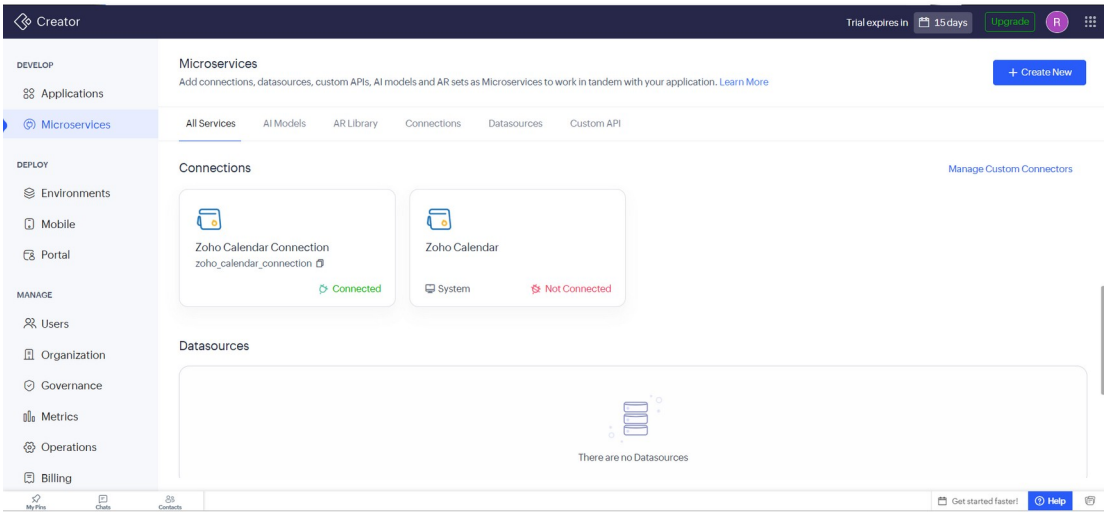


Figure 5.29: Connections List Showing the Zoho Calendar Connection

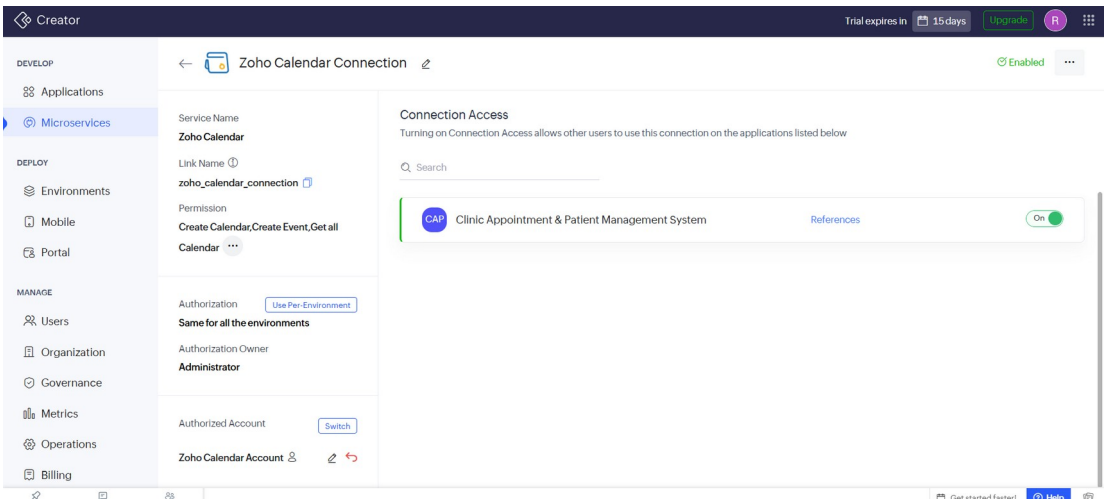


Figure 5.30: Connection Details and Access Turned On for the Application

The calendar identifier used by the script was taken from the Zoho Calendar settings: under My Calendars → Integrate Calendar the page shows the CalDAV URL, the Calendar ID and the Calendar UID, and the Calendar UID (a5dea53e…) is the value passed to zoho.calendar.createEvent in Code 5.1 (Figure 5.31). With the connection, the identifier and the script in place, saving an appointment sends the event details to Zoho Calendar for the visit to be added to the clinic's calendar.

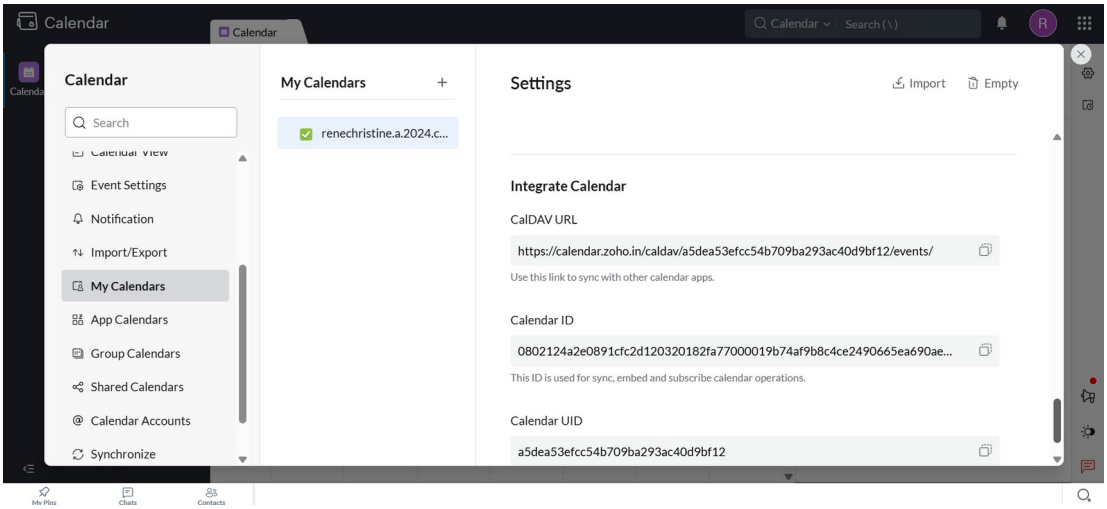


Figure 5.31: Zoho Calendar Settings Showing the Calendar UID

Table 5.7: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Calendar scheduling	Zoho Calendar – createEvent	Creator connection zoho_calendar_connection + Deluge zoho.calendar.createEvent	Zoho connection authorised by the Administrator	Event titled “Patient Appointment” with start and end time (60 minutes) taken from the appointment

No REST call with invokeurl, webhook or Creator REST API was used, because the connection covers the only external service the application needs.

5.8 User Management and Role-Based Access

Access to the application is controlled under Settings → Users, which has three tabs: Permissions, Roles and Data Sharing.

Permission sets. The Permissions tab lists the two built-in sets Read and Write (“This profile will have read / write permission for all components”) and four sets created for the clinic: Accountant (“Manage billing and payments”), Admin (“Full access to all clinic modules”), Doctor (“Manage treatments and view appointments”) and Receptionist (“Register patients and schedule appointments”) (Figure 5.32). Each set decides which forms, reports and pages a user may open, view, edit or delete, in line with the matrix of Table 4.7.

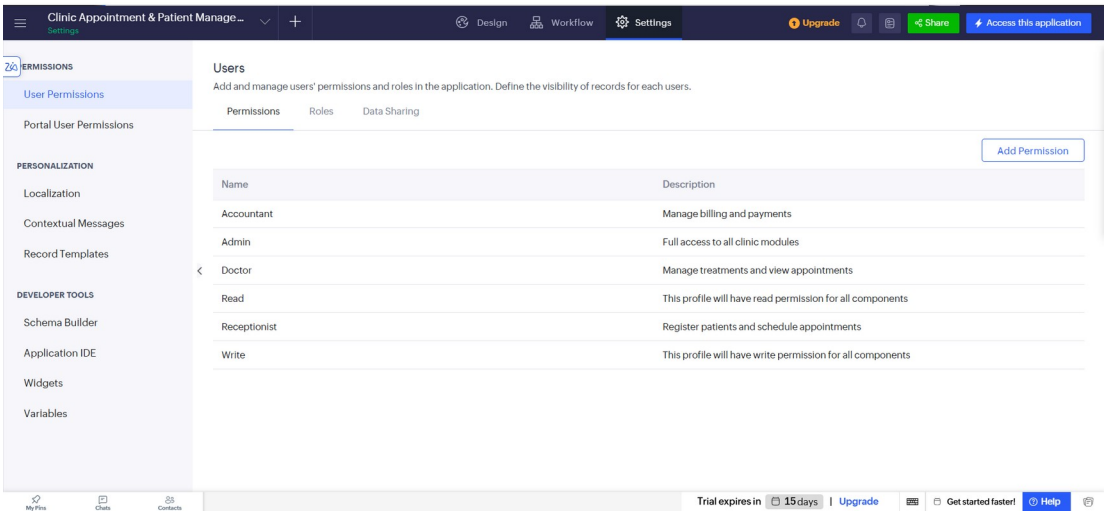


Figure 5.32: Permission Sets: Accountant, Admin, Doctor, Read, Receptionist and Write

Roles. The Roles tab holds the hierarchy CEO → Admin → Doctor, Receptionist and Accountant (Figure 5.33). The CEO is at the top, the Admin role is below it, and the Doctor, Receptionist and Accountant roles are at the same level under Admin. A user in a higher role can see the records of the roles below it.

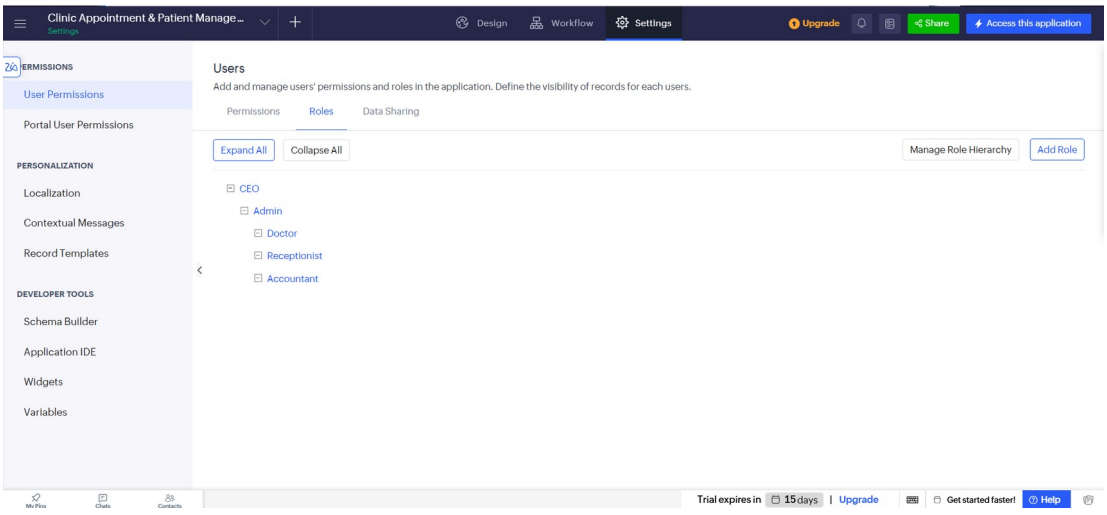


Figure 5.33: Role Hierarchy: CEO → Admin → Doctor, Receptionist, Accountant

Data sharing rules. The Data Sharing tab holds two rules, both with the level of access Read Only (Figure 5.34): one shares records of the Appointments form from the role Receptionist to the role Doctor, so that a doctor can see the appointments booked by the front desk, and the other shares records of the Billing form from the role Admin to the role Accountant, so that the accountant can see the bills.

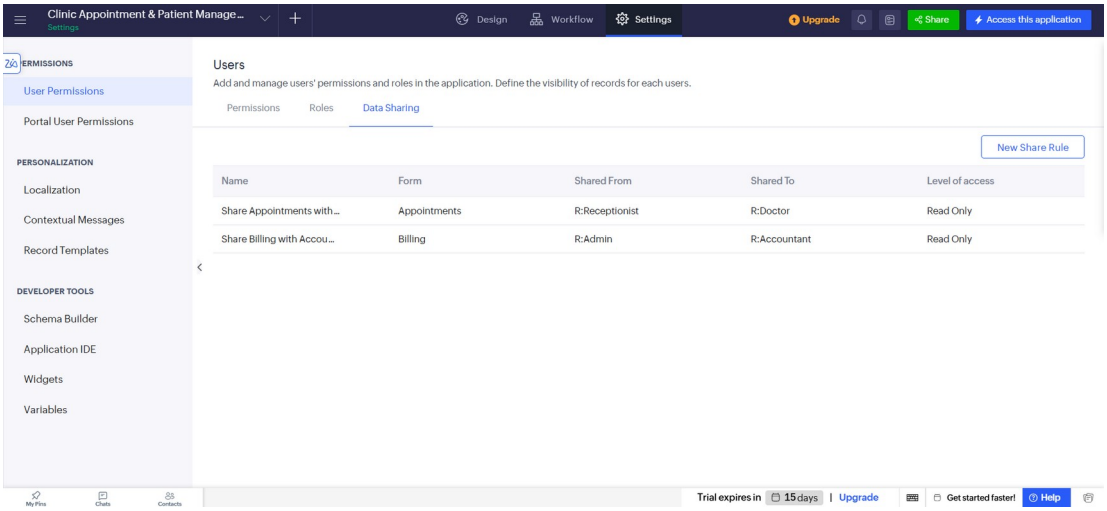


Figure 5.34: Data Sharing Rules of the Application

Table 5.8: Permission Sets and Roles

Name	Permission set description	Position in role hierarchy
Admin	Full access to all clinic modules	Below CEO; parent of Doctor, Receptionist and Accountant
Doctor	Manage treatments and view appointments	Under Admin
Receptionist	Register patients and schedule appointments	Under Admin
Accountant	Manage billing and payments	Under Admin
Read / Write (built-in)	Read or write permission for all components	Not part of the hierarchy

Table 5.9: Data Sharing Rules

Rule	Form	Shared From	Shared To	Access Level
Appointments shared with Doctor	Appointments	Receptionist	Doctor	Read Only
Billing shared with Accountant	Billing	Admin	Accountant	Read Only

The permission sets, role hierarchy and data sharing rules are saved and listed in Settings → Users. Their effect for each role is checked by signing in as a user of that role during testing (Chapter 6).

CHAPTER 5
IMPLEMENTATION (continued)

5.9 AI and Analytics Features

Analytics. The analytics in the application are on the Clinic Dashboard (Figure 5.35). A bar chart, Appointments by Date, counts appointments per month; at the time of the screenshot it showed one appointment in October 2026 and three in November 2026. Below it, two embedded list reports show the Upcoming Appointments (3 of 3 records) and the Pending Bills (4 of 4 records). A circular chart above the bar chart is partly visible at the top of the screenshot. Zoho Analytics was not integrated.

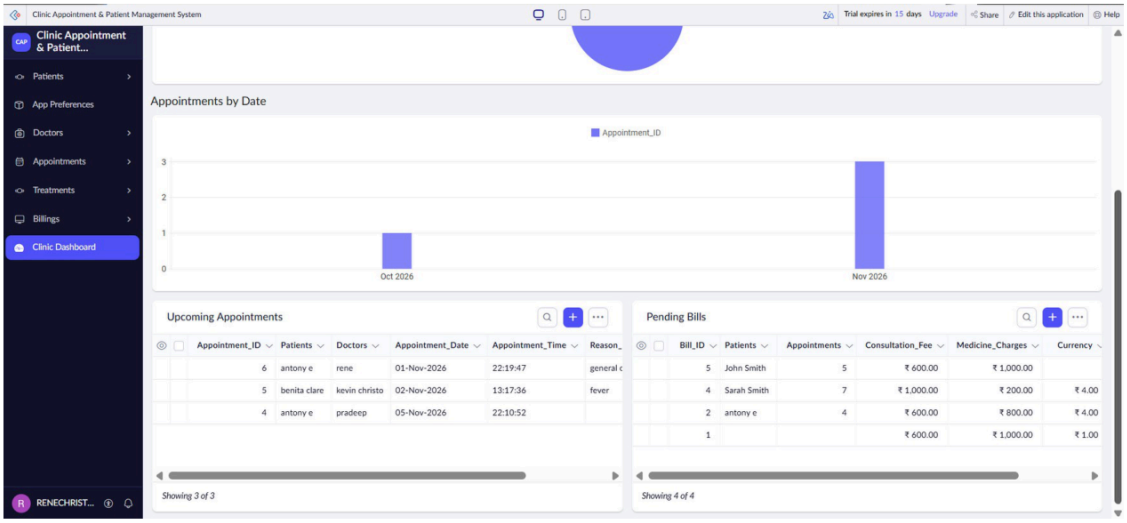


Figure 5.35 **:** Clinic Dashboard – Appointments by Date Chart, Upcoming Appointments (3 of 3) and Pending Bills (4 of 4)

AI features of the platform. No Zia or other AI feature of Zoho Creator (AI fields, prediction, OCR, keyword extraction, sentiment analysis or the AI application builder) is part of the delivered application. The Zia icon is visible in the builder and application screens but was not used in the application.

AI assistance during the project. An AI assistant (Claude, by Anthropic) was used to help draft this report from the screenshots of the application. All configuration of the application in Zoho Creator, including the forms, the dashboard, the permission sets and the Deluge script, was carried out by the student, and the results were checked against the live application as shown in Chapter 6.

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst, custom functions, JavaScript widgets or the Creator REST API) was used, because the required behaviour (storing patient, doctor, appointment, treatment and billing records, a validation script and a dashboard) was covered by forms, reports and one short Deluge script. A future extension could expose the Appointment form through the Creator REST API so that a website or mobile app could book appointments online, or use a Catalyst function to send SMS appointment reminders and to run heavier processing such as billing analytics.

5.11 Mobile and Portal Access

The application header offers desktop, tablet and phone preview toggles (visible in Figures 5.1 and 8.3), but a dedicated mobile configuration, the Zoho Creator mobile app and a patient portal were not set up or tested. Letting patients book appointments and view their bills through a portal is listed as a future enhancement (Section 9.2).

Considerations for mobile and patient-facing access

A mobile interface can be useful at the clinic because reception staff may need to check the day's schedule while away from the main workstation. Before relying on the mobile preview for routine work, the application should be checked on the screen sizes and devices used by the clinic. The checks should include form readability, lookup selection, date and time entry, report filtering, and whether long text fields can be completed comfortably. The current project demonstrates the application structure and responsive preview options; it does not claim that a separate mobile deployment has been validated.

A patient portal would require additional design decisions beyond exposing the existing internal forms. Patient identity must be verified, and each patient should be able to access only their own appointments, treatment information and bills. The clinic would also need to decide which fields patients may edit, how appointment changes are approved, and how cancelled or rescheduled visits are communicated. These controls should be tested with dedicated patient accounts before portal access is enabled. For this reason, portal booking and bill viewing remain proposed enhancements, not completed functions of the present application.

For any future mobile or portal release, a short acceptance checklist should be completed on the target devices and accounts. It should cover successful sign-in, access restrictions, form submission, error messages, report visibility and the handling of an interrupted network connection. Results should be recorded with the device, account role, date and outcome so that the release decision is supported by repeatable evidence.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done on the live application in Zoho Creator while it was being built. **Unit testing** checked each form by entering sample data (three patients, appointments, treatments and bills) and each report and chart by comparing it with the records. **Validation testing** checked the Patient Data Validation script by submitting a new patient with a phone number that already belonged to a saved patient. **Role-based testing** by signing in as users of each permission set (Admin, Receptionist, Doctor, Accountant) was planned but is not recorded; the permission sets were only checked for saving correctly (Section 5.8). **User acceptance testing** with other users has not been recorded yet (Section 6.5). The sample appointments and bills include dates and amounts entered for testing and are not real clinic data.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a patient with all fields	Name “John Smith”, date of birth, phone +91 7397476603, e-mail, address, blood group A+, gender male; Submit	Record saved	Patient listed in All Patients (3 of 3) (Fig. 8.1, 8.3)	Pass
TC-02	FR-01	Duplicate phone number rejected	Enter a new patient with phone +91 7397476603, already held by a saved patient; Submit	Alert and submission cancelled	Alert “Phone Number already exists”; record not saved (Fig. 6.1)	Pass
TC-03	FR-01	Full Name mandatory	Leave first and last name empty; Submit	Alert “Full Name is required”	Not executed	Not run
TC-04	FR-01	Phone Number mandatory	Leave phone number empty; Submit	Alert “Phone Number is required”	Not executed	Not run
TC-05	FR-01	Editing a patient keeps own phone number	Edit a saved patient without changing the phone number	Edit allowed (own record ignored)	Not executed	Not run
TC-06	FR-02	Book an appointment with Patient and Doctor lookups	Choose patient, doctor, date, time and reason	Record saved	Appointments 4, 5 and 6 listed in Upcoming Appointments (3 of 3) (Fig. 5.17 , Fig. 5.21	Pass
TC-07	FR-03	Record a treatment	Choose patient and appointment; enter treatment details	Record saved	No treatment record appears in the screenshots	Not verified
TC-08	FR-04	Create a bill for an appointment	Enter consultation fee and medicine charges	Record saved	Bills 1, 2, 4 and 5 listed in Pending Bills (4 of 4) (Fig. 5.23 , 6.3)	Pass
TC-09	FR-05	Dashboard chart counts appointments by month	Open Clinic Dashboard	One bar per month	Oct 2026: 1; Nov 2026: 3 ( )	Pass
TC-10	FR-06	Permission sets can be saved	Create Accountant, Admin, Doctor and Receptionist sets	Saved and listed	All four listed with Read and Write (Fig. 7.3)	Pass
TC-11	FR-06	Receptionist cannot see billing	Sign in as a Receptionist user	Billing not offered	Not executed – no sign-in test recorded	Not run
TC-12	FR-07	Publish to Stage and Production	Publish from Environments	Version listed in both	Version 2.0 in Stage and Production, Nov 01, 2026 (Fig. 7.1)	Pass

FR references: FR-01 patient registration and validation; FR-02 appointment booking; FR-03 treatment records; FR-04 billing; FR-05 dashboard and reports; FR-06 permission sets; FR-07 deployment. Match these numbers with the requirement table of Chapter 3.

6.3 Debugging Deluge Scripts

Debugging relied on three techniques: the alert message shown by the form on submission (Figure 6.1), the behaviour of the live records after a test submission (a rejected submission leaves the number of records unchanged, Figure 8.3), and the line-by-line view of the script in the Deluge editor (Figure 6.2). The script is saved as an action of the workflow Patient Data Validation, which runs on Patients → Created or Edited → Validations on form submission, and the editor accepted it without a syntax error.

A design point handled in the script. A duplicate check that compares every saved record with the entered phone number would also reject the edit of an existing patient, because the patient's own record matches. The script therefore ignores the record being edited by comparing record IDs (rec.ID != input.ID) and applies the check to every record when the form is new (input.ID == null).

Deliberate omission. The script ends with the comment "No Email Validation". E-mail is not checked, and the three sample patients share one address (Figure 8.3); this is a decision and not a defect.

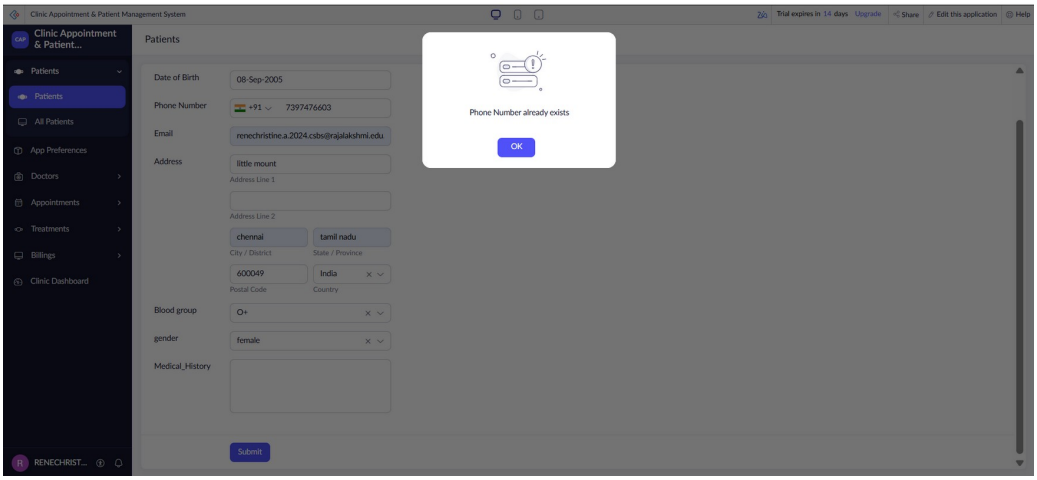


Figure 6.1: Patients Form Showing the Alert "Phone Number already exists" after Submit

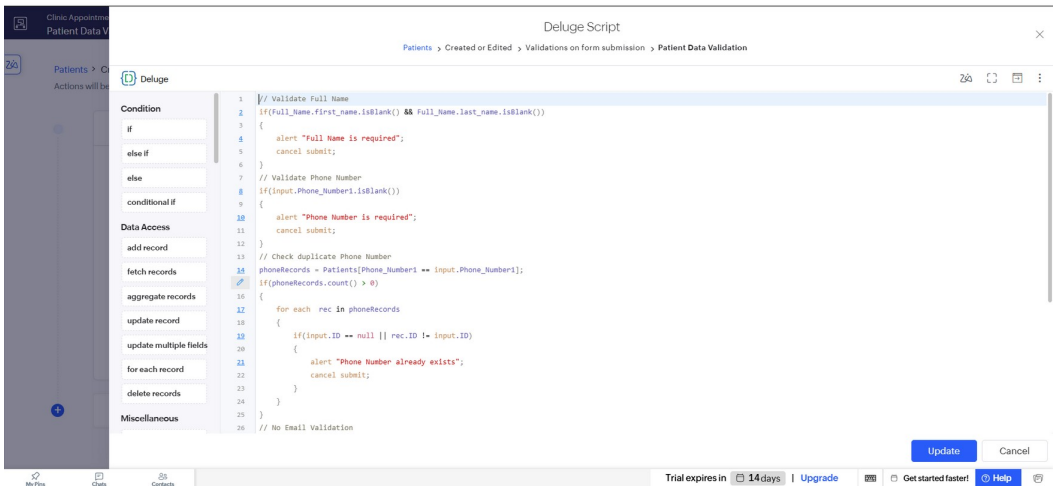


Figure 6.2: Patient Data Validation Script in the Deluge Editor (Patients → Created or Edited → Validations on form submission)

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	A new patient could be saved with a phone number that already belongs to another patient.	No uniqueness check on the phone number.	Patient Data Validation script fetches the patients with the same phone number and cancels the submission (Fig. 6.1, 6.2).	Closed
BUG-02	The duplicate check would also block the edit of a saved patient.	The patient's own record matches the phone number.	Script skips the record whose ID equals the edited record (rec.ID ! = input.ID).	Closed
BUG-03	The Environments page showed "Waiting for development app creation" and "Creating development app" instead of a version.	The development application was still being created after the first publish.	Waited; Stage and Production then showed version 2.0 (Fig. 7.1, 7.2).	Closed
BUG-04	Pending Bills lists Bill 1 with an empty Patients and Appointments column (Fig. 6.3).	Bill saved without the Patient and Appointment lookups; not made mandatory.	None yet; make both lookups mandatory.	Open
BUG-05	Pending Bills shows the patient "Sarah Smith", who is not among the 3 patients in All Patients (Fig. 6.3, 8.3).	Not determined from the screenshots (name typed in a different form or record removed).	None yet; check the Patient lookup of the bill.	Open
BUG-06	The Currency column of Pending Bills holds small amounts (₹4.00, ₹1.00) and is empty for Bill 5 (Fig. 6.3).	Purpose of the field is unclear; it sits beside the fee columns.	None yet; rename the field or correct the values.	Open
BUG-07	The three sample patients share one e-mail address and e-mail is not validated.	The script has no e-mail check by design.	Accepted; an e-mail format check is proposed.	Accepted
BUG-08	John Smith's date of birth is 03-Apr-2002 on the form (Fig. 8.1) and 04-Apr-2007 in All Patients (Fig. 8.3).	The record may have been edited after the first entry; not determined.	None yet; check the record history.	Open

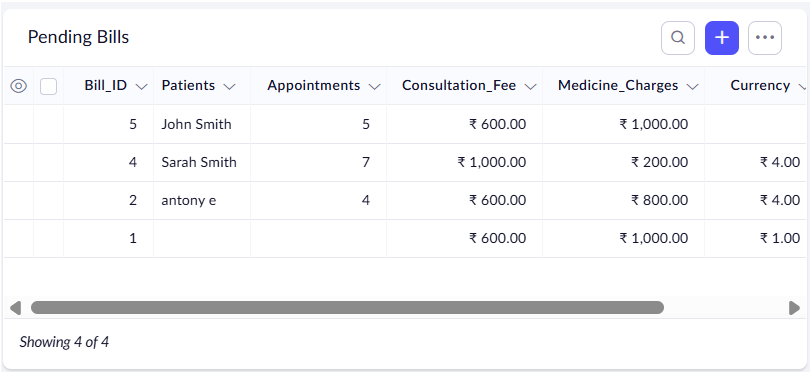


Figure 6.3: Pending Bills Report (4 of 4) Showing Bill 1 without a Patient and the Currency Column (BUG-04 to BUG-06)

6.5 User Acceptance Testing

No feedback from other users has been recorded for this application yet. Table 6.3 lists the users who are to test it and is to be completed with their feedback after the trial.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Front-desk staff member	Receptionist	Not yet collected	To be recorded
Clinic doctor	Doctor	Not yet collected	To be recorded
Accounts staff member	Accountant	Not yet collected	To be recorded

6.6 Additional Deluge Validation Evidence

The following screenshots document the form-submission validation scripts configured in Zoho Creator. They supplement the patient duplicate-phone validation and test cases discussed in Sections 6.1-6.4.

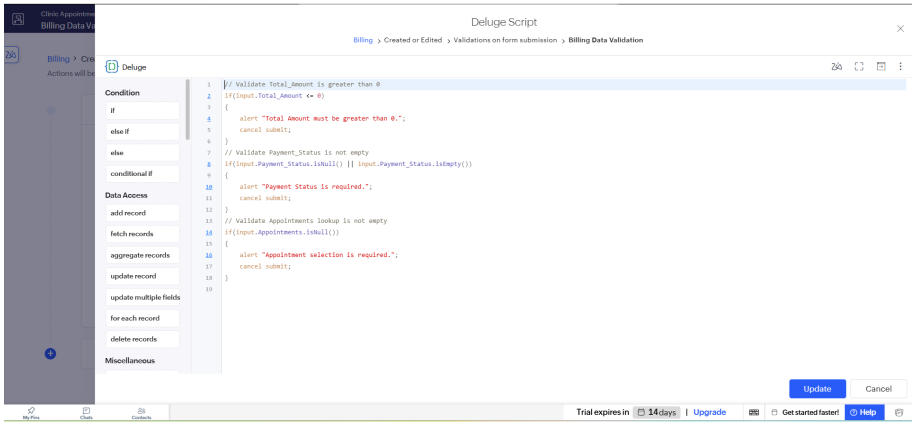


Figure 6.4: Billing Data Validation - total amount, payment status and appointment lookup checks.

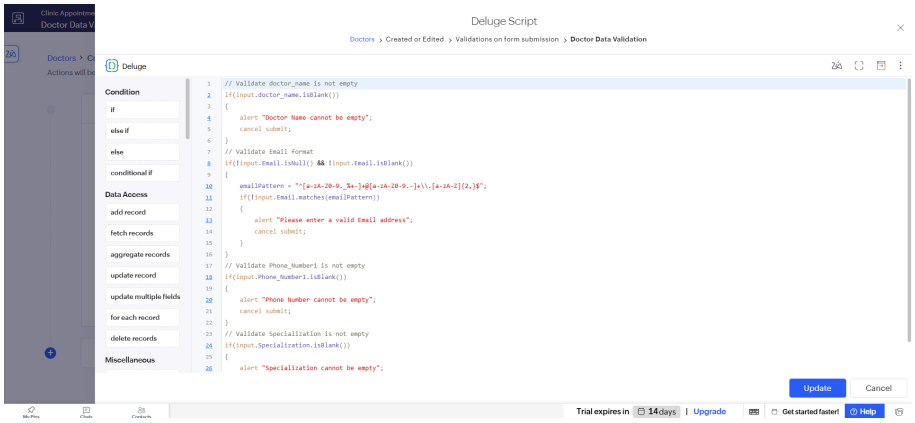


Figure 6.5: Doctor Data Validation - required doctor name, valid e-mail format, phone number and specialization.

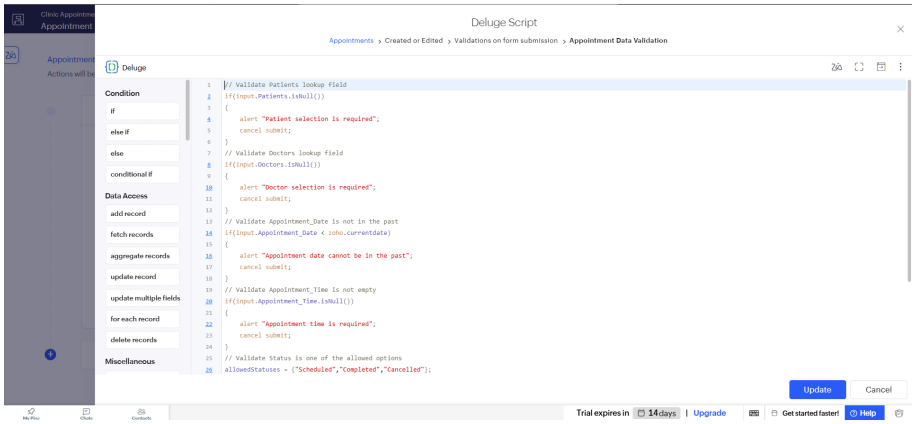


Figure 6.6: Appointment Data Validation - patient and doctor lookups, non-past date, time and permitted status.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides a multi-stage deployment space (Deploy → Environments) with Stage and Production environments, and development takes place in the builder. After the first publish, the Environments page listed the Clinic Appointment & Patient Management application with the status "Creating development app", the text "Waiting for development app creation" in Stage and version 1.0 in Production (Figure 7.2). Once the development application had been created, the page listed version 2.0 in Stage and version 2.0 in Production, both dated Nov 01, 2026, with the note "No changes available" (Figure 7.1). The page offers the buttons Version History, Manage and Publish.

The rollback plan is to keep the previous version in the Version History of the Environments page and to republish it if a later change breaks the application; no rollback was exercised.

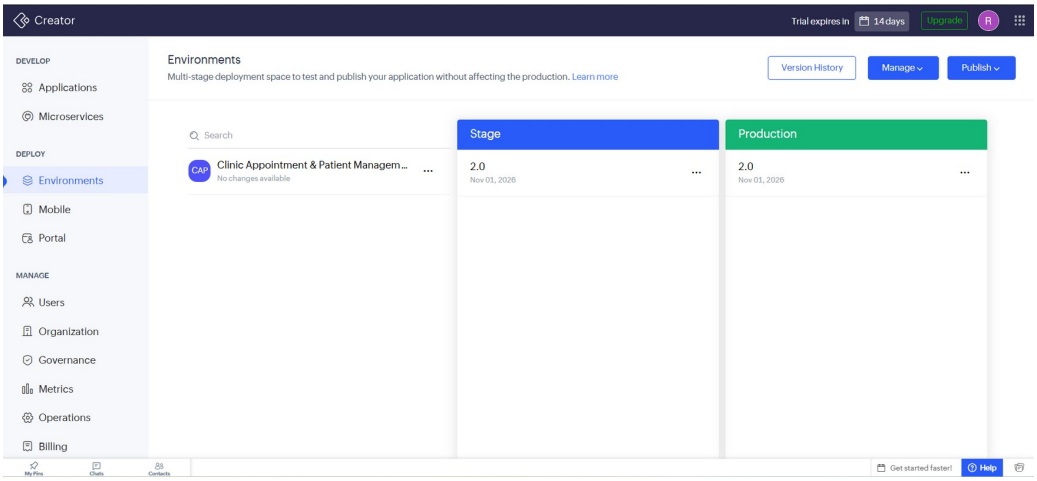


Figure 7.1: Environments Page Listing the Application in Stage (2.0, Nov 01, 2026) and Production (2.0, Nov 01, 2026)

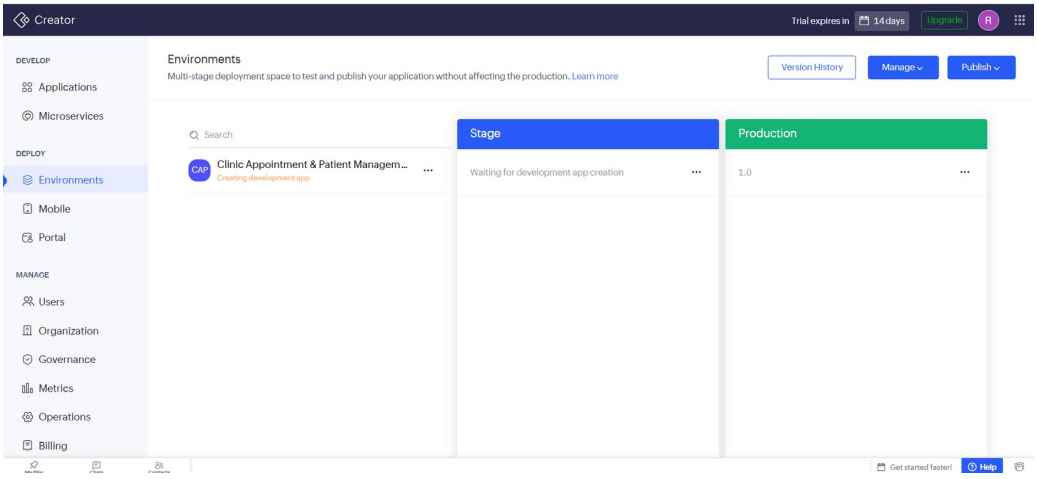


Figure 7.2: Environments Page Showing "Waiting for development app creation" in Stage and Version 1.0 in Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (·)
1	All test cases executed in Development / Stage. (TC-03, TC-04, TC-05, TC-07 and TC-11 not run or not verified.)	✗
2	Sample/test data removed or isolated. (Sample patients, appointments and bills remain.)	✗
3	Roles, permissions and users configured in Production. (Four permission sets defined; no sign-in test with users recorded.)	✗
4	Connections authorised in Production. (No external connection is used.)	Not applicable
5	Schedules and workflows enabled. (The form workflow with the validation script is in use; no schedules exist.)	✓
6	Application published and URL verified. (Stage 2.0 and Production 2.0 listed; URL check not recorded.)	Partial
7	Version / release notes recorded. (Not recorded.)	✗

7.2 Performance Optimisation

No performance measurements were taken, so no improvement is claimed. The design choices that help performance are: the dashboard shows two short lists (Upcoming Appointments and Pending Bills) instead of every record; the appointment chart counts appointments per month instead of plotting each record; and the validation script runs a single fetch with a criteria on the phone number, so the loop only visits records that already match. Table 7.2 records this honestly.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Aggregate count by month	Appointments by Date chart	Not measured	Two bars (Oct 2026, Nov 2026)	Readable chart; load time not measured
Criteria-based fetch in the script	Patient Data Validation	Not measured	Only records with the same phone number are visited	Fewer records processed; time not measured
Short lists on the dashboard	Upcoming Appointments, Pending Bills	Not measured	3 and 4 records shown	Fewer rows; load time not measured

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Permission sets	Accountant (manage billing and payments), Admin (full access to all clinic modules), Doctor (manage treatments and view appointments), Receptionist (register patients and schedule appointments), plus the built-in Read and Write sets (Fig. 7.3)	Users reaching modules that are not part of their work
Mandatory-field validation	Full Name and Phone Number checked in the Deluge script	Incomplete patient records
Duplicate phone check	Patients form, Created or Edited	Duplicate patient records

Security Control	Where Applied	Threat Mitigated
Stage and Production environments	Deploy → Environments (Fig. 7.1)	Untested changes reaching live data
Account sign-in	Zoho Creator login	Anonymous access to patient data

Not applied: field-level encryption, the personal-data and health-information flags on fields such as Medical_History, IP or domain restrictions, two-factor authentication and record-level criteria. The Deluge script contains no passwords or API keys. Because the Patients form stores medical history and contact details, these controls should be added before real patient data is used.

7.4 Governance and Administration

Administration is done in Settings (Users → Permissions, Roles, Data Sharing) and in the Environments page. Naming follows a simple convention: one menu section per entity (Patients, Doctors, Appointments, Treatments, Billings), reports named after their purpose (All Patients, Upcoming Appointments, Pending Bills) and the validation workflow named after what it does (Patient Data Validation). A dedicated audit trail, change history and backup/export configuration was not evidenced, and the application has no named maintainer other than the administrator who owns it. The header shows that the account is on a trial edition (“Trial expires in 14 days”, with an Upgrade link), so the application must be moved to a paid plan before long-term use. Figure 7.3 shows the permission sets in the administration page.

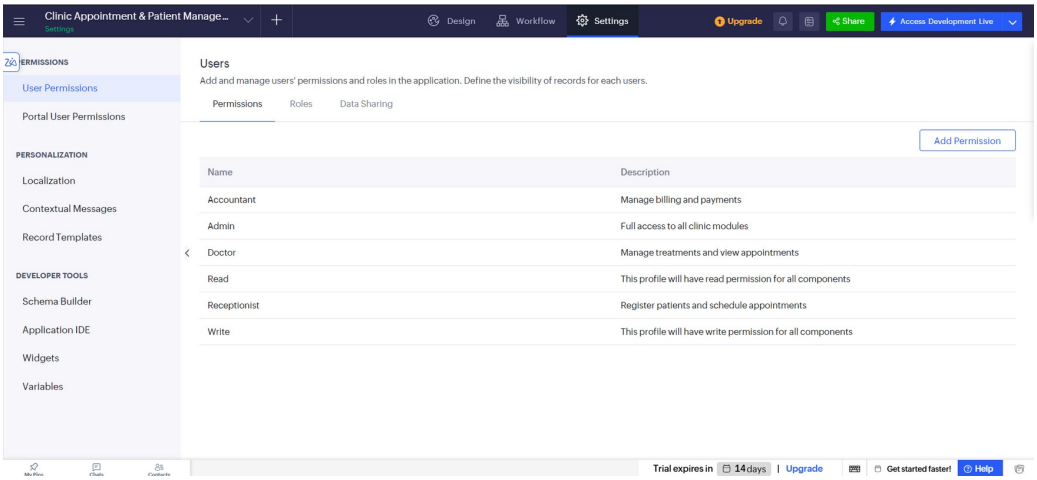


Figure 7.3: Settings → Users → Permissions Listing Accountant, Admin, Doctor, Read, Receptionist and Write

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application is presented through one scenario that shows the patient validation. In step 1 the first patient, John Smith, is entered on the Patients form with his name, date of birth, phone number +91 7397476603, e-mail, address, blood group A+ and gender male (Figure 8.1). In step 2 a different patient (female, blood group O+, Chennai 600049) is entered with the same phone number; on Submit the Patient Data Validation script finds the saved patient with that number, shows the alert “Phone Number already exists” and cancels the submission (Figure 8.2). In step 3 All Patients shows three records (antony e, benita clare and John Smith), so no duplicate was stored (Figure 8.3). In step 4 the Clinic Dashboard shows the appointments by month, the three upcoming appointments and the four pending bills (Figure 8.4).

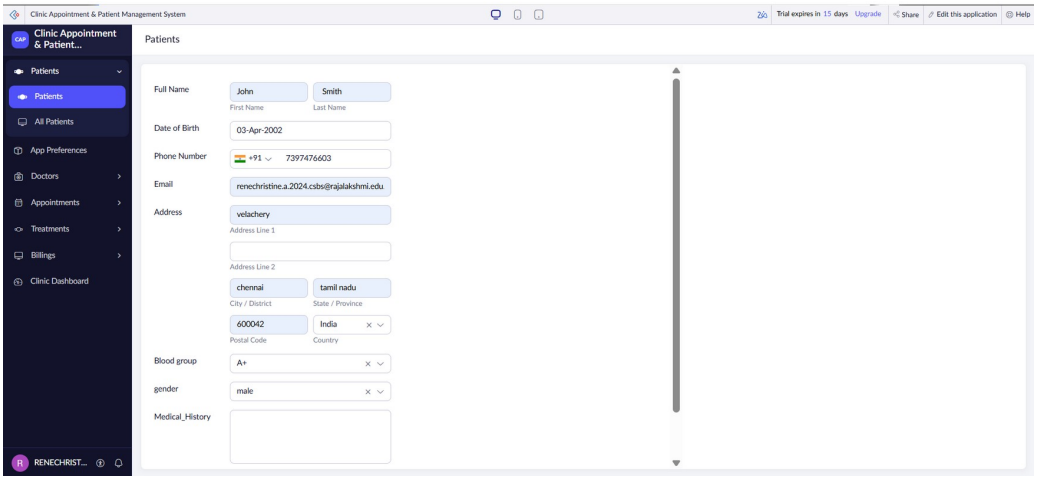


Figure 8.1: Scenario Step 1 – Patients Form Filled for John Smith (Phone +91 7397476603, Blood Group A+, Gender Male)

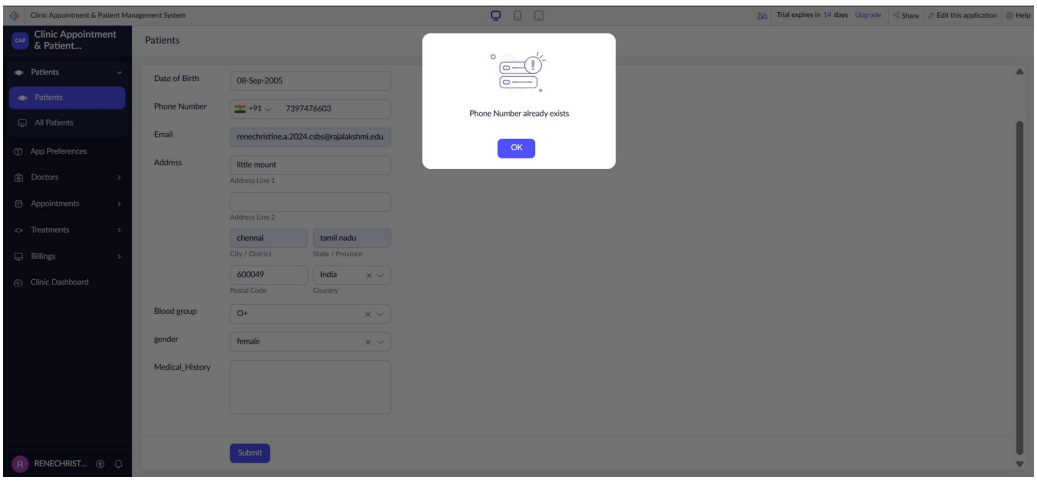


Figure 8.2: Scenario Step 2 – Alert “Phone Number already exists” When the Same Phone Number Is Entered Again

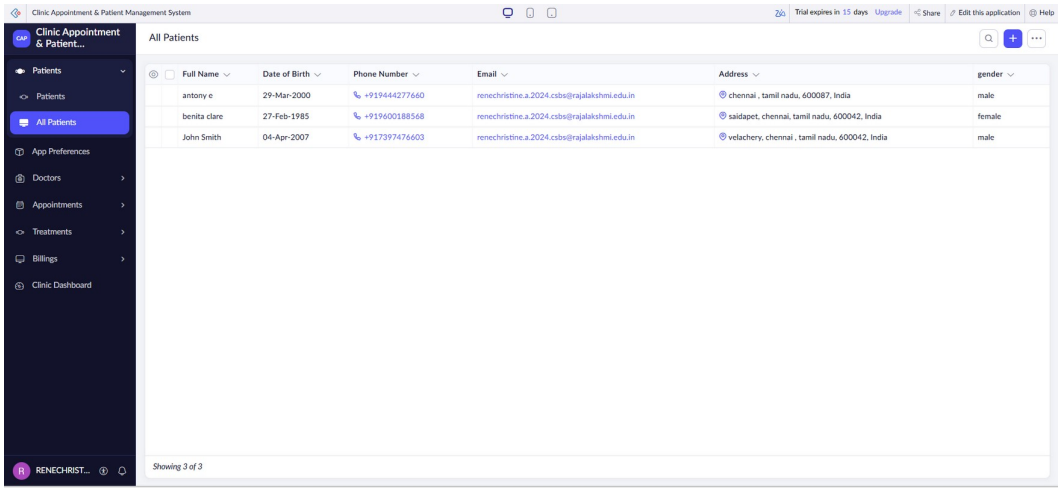


Figure 8.3: Scenario Step 3 – All Patients Report Showing 3 of 3 Records

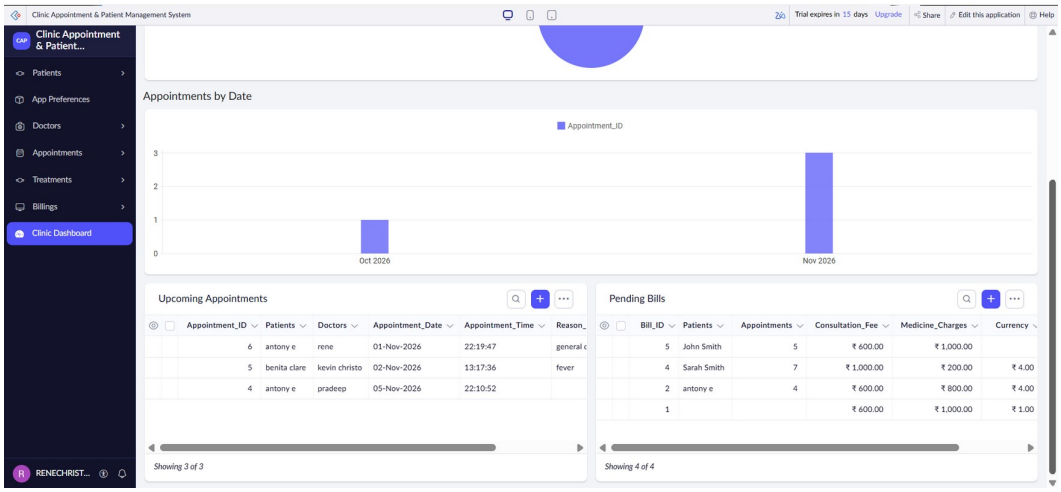


Figure 8.4: Scenario Step 4 – Clinic Dashboard with Appointments by Date, Upcoming Appointments and Pending Bills

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. To store patient, doctor, appointment, treatment and billing records in linked forms	Sections 4.2, 4.3, 5.2	Achieved. Menu sections for all five entities; patients, appointments and bills stored (Fig. 8.3, 5.1).
2. To validate patient data with a Deluge script	Sections 5.5, 6.3	Achieved. Full Name and Phone Number are checked and a duplicate phone number is rejected (Fig. 6.1, 6.2).
3. To provide reports and a dashboard	Sections 5.3, 5.4, 5.9	Achieved. All Patients report, Appointments by Date chart, Upcoming Appointments and Pending Bills (Fig. 8.3, 8.4).
4. To control access through permission sets	Sections 5.8, 7.3	Partly achieved. Accountant, Admin, Doctor and Receptionist sets defined (Fig. 7.3); sign-in tests with users not recorded.
5. To deploy through Stage and Production	Section 7.1	Achieved. Version 2.0 listed in Stage and Production (Fig. 7.1).

8.3 Comparison with the Existing System

No timing or error-rate measurements were made for either the manual process or the application, so Table 8.2 compares them qualitatively and no percentage improvements are claimed.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Patient registration	Paper register or cards; structure varies	One form with name, date of birth, phone, e-mail, address, blood group, gender and medical history	Consistent, complete records
Duplicate patients	Found only by chance	Duplicate phone number rejected with an alert	Duplicates blocked at entry
Appointment scheduling	Phone calls and a diary	Appointment records linked to patient and doctor	Appointments stored against the right patient and doctor
Billing follow-up	Counting unpaid slips by hand	Pending Bills report on the dashboard	Immediate list of pending bills
Summary reporting	Counting by hand	Appointments by Date chart	Counts per month on one page
Data accessibility and control	Whoever holds the register sees everything	Permission sets for Admin, Receptionist, Doctor and Accountant (defined, not tested with sign-ins)	Controlled access once users are assigned

Table 8.3 records the quantities that can be read from the screenshots. They describe the sample data at the time of the screenshots and are observations, not performance measurements.

Table 8.3: Observed Application State from the Screenshots

Measure	Observed Value	Source
Patients in All Patients	3 of 3	Figure 8.3
Appointments in October 2026	1	Figure 8.4
Appointments in November 2026	3	Figure 8.4
Upcoming Appointments shown	3 of 3 (IDs 4, 5, 6; 01, 02 and 05 Nov 2026)	Figure 8.4
Pending Bills shown	4 of 4 (Bill IDs 1, 2, 4, 5)	Figure 6.3
Sum of consultation fees in Pending Bills	₹2,800 (600 + 1,000 + 600 + 600)	Figure 6.3
Sum of medicine charges in Pending Bills	₹3,000 (1,000 + 200 + 800 + 1,000)	Figure 6.3
Pending bills without a patient	1 (Bill 1)	Figure 6.3
Duplicate phone number submitted	Rejected with alert	Figure 8.2

8.4 Limitations of the Low-Code Approach

The following constraints and issues were met while building this application.

- **Editions and trial limits.** The application was built on a trial edition (“Trial expires in 14 days”), so user invitation and some governance features could not be assumed to be available.
- **Validation scope.** The script checks only Full Name, Phone Number and duplicate phone numbers. E-mail format, date-of-birth range and blood-group rules are not checked, and a name is accepted if either the first or the last name is filled.

- **Lookup data quality.** Lookups do not force a value; a bill can be saved without a patient or appointment (BUG-04).
- **Reporting depth.** Standard reports and charts count and list records; revenue totals or doctor-wise workload would need extra aggregate reports or Zoho Analytics.
- **Platform dependency.** The application exists only inside Zoho Creator, and the logic is written in Deluge, which is specific to Zoho (vendor lock-in).
- **Scripting scope.** Deluge covered every need of this project, but its statement and fetch limits were not tested because the script is small.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code configuration: the forms and their lookups, the reports, the chart, the dashboard page, the permission sets and the Stage/Production publishing. Low-code was needed for one short Deluge script, the Patient Data Validation. Pro-code was not needed. A move to pro-code would be justified if the system had to take bookings from a website or mobile app through the Creator REST API, to send SMS reminders through an external gateway, or to integrate with a laboratory or insurance system, because these go beyond what forms, reports and a short script can do comfortably.

Additional Output Evidence · Appointment Confirmation

Figure 8.5 shows the appointment confirmation email generated by the application. The message states that the appointment was booked successfully and displays the appointment date as 31-Oct-2026 and the appointment time as 22:32:34. The email provides visible evidence that the appointment confirmation notification was generated and received in the mailbox. It supports the appointment-booking notification result; it does not, by itself, verify calendar-event creation or role-based access.

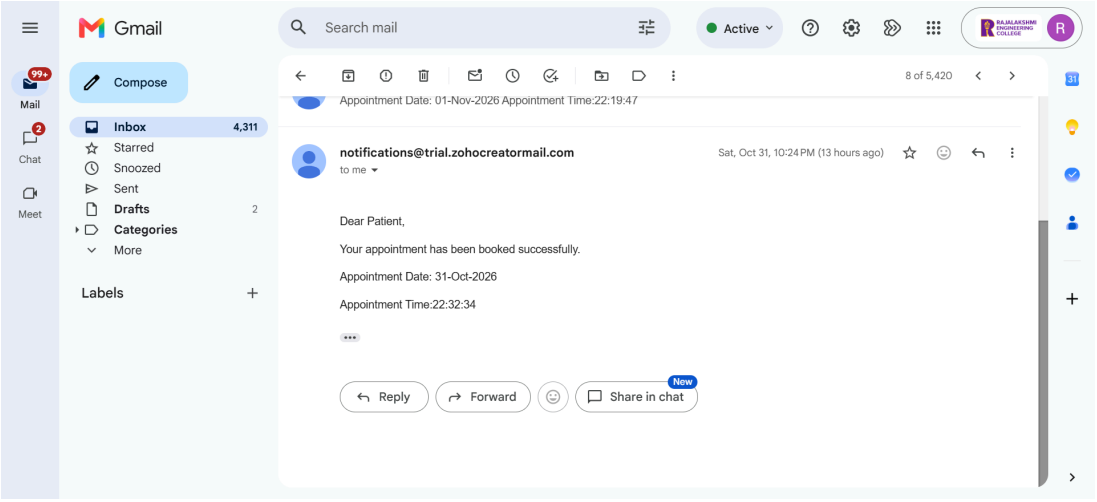


Figure 8.5: Gmail Confirmation Email Showing Successful Appointment Booking

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Clinic Appointment & Patient Management System shows how a paper-based clinic routine can be replaced by a small low-code application. Linked forms store patients, doctors, appointments, treatments and bills. A Deluge validation script requires a name and a phone number and rejects duplicate phone numbers. The Clinic Dashboard shows the appointments by month, the upcoming appointments and the pending bills, and permission sets for the Admin, Receptionist, Doctor and Accountant define what each kind of user may do. The application was published to the Stage and Production environments as version 2.0.

The work also has limits that are recorded openly: sign-in tests with users of each permission set are not recorded, one bill was saved without a patient, e-mail is not validated, no timing measurements were made, and feedback from other users has not yet been collected. These items are the first to address before the application is used for real patient records.

9.2 Future Enhancements

- Invite users, assign them to the Admin, Receptionist, Doctor and Accountant permission sets, add record-level criteria so that each user sees only what the work needs, and test every role.
- Make the Patient and Appointment lookups mandatory in Billings and correct the bills that have none (BUG-04 to BUG-06).
- Add e-mail format validation, a date-of-birth range check and a rule that stops a doctor from being booked twice at the same time.
- Add a scheduled workflow that sends appointment reminders to patients, and an e-mail or SMS confirmation after a booking.
- Add an approval process or Blueprint for the billing cycle (appointment → treatment → bill → payment).
- Publish a patient portal or use the Creator mobile app so that patients can book appointments and view their bills.
- Flag Medical_History as health information, encrypt sensitive fields and turn on an audit trail.
- Integrate Zoho Analytics for revenue and doctor-wise reports.

9.3 Learning Outcomes and Career Relevance

Building this application gave practice in the main parts of Zoho Creator: data modelling with forms and lookups, validations, list reports and chart reports, UI Builder pages, form workflows, Deluge scripting with fetch and loops, permission sets, and deployment through environments. It also showed how testing and debugging work in a low-code platform, including the cases where a fix is not found immediately. These skills relate to the roles of citizen developer, low-code developer, solutions

consultant and automation engineer, and they provide a basis for moving to pro-code extensions when requirements outgrow the platform.

Applying the learning outcomes

The project provided practice in translating a business process into a structured application. Defining the Patients, Doctors, Appointments, Treatments and Billing forms required attention to the meaning of each field, the records that depend on it, and the points at which a user must supply information. Lookup fields demonstrated how related records can be reused instead of repeatedly entering the same patient or doctor details. Reports and dashboard components then provided different views of the same stored data for operational use.

The Deluge work introduced event-driven validation and workflow automation. The patient validation example illustrates why a rule must consider both record creation and record editing: a duplicate phone number should be rejected for a new patient, but an unchanged number should not prevent that patient's existing record from being edited. The billing and appointment validation scripts also show how clear alerts and cancellation of submission can guide users toward complete records. These scripts should continue to be tested with valid, invalid and boundary-value inputs whenever fields or workflows are changed.

The project also highlighted the importance of evidence-based reporting. A feature being visible in the builder does not by itself prove that it works for every role or in every environment. Screenshots, saved records, test cases and defect logs provide different kinds of evidence, and open items must remain visible until they are retested and closed. This discipline is relevant to future work in application development, business analysis, software testing and workflow automation.

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Only sources cited in the report are listed.

APPENDIX A
ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1 lists the main Zoho Creator features and records whether each one was used in this application, with the section of the report that gives the evidence. “Used” means the feature is configured and shown in the report; “Used – not verified” means it is configured but its execution was not evidenced; “Not used” means it was deliberately left out of the delivered scope.

Table A.1: Feature Coverage Checklist

S.No.	Zoho Creator feature	Status	Evidence / remark
1	Forms (5 forms: Patients, Doctors, Appointments, Treatments, Billing)	Used	Sections 4.3, 5.2
2	Subforms	Not used	5 forms / 0 subforms (Project at a Glance)
3	Field types – Auto Number, Name, Single Line, Multi Line, Date, Time, Phone, Email, Address, Drop Down, Lookup, Currency	Used	Section 5.2 (twelve field types)
4	Lookup fields (one-to-many relationships)	Used	Section 4.2, Table 4.1
5	Lookup shown as radio buttons	Used	Section 5.2.4, Figure 5.9
6	List reports (auto-created and custom)	Used	Section 5.3, Table 5.3
7	Filtered reports (Today’s, Upcoming, Pending Bills)	Used	Section 5.3.2, 5.3.4
8	Calendar report (Appointment Calendar)	Used – not verified	Listed in the menu (Table 5.1); no screenshot of the calendar view
9	Charts on a page (pie and bar)	Used	Section 5.4, Figure 5.23
10	Pages built in the UI Builder (Clinic Dashboard)	Used	Section 5.4, Table 5.4
11	Summary cards and embedded reports on a page	Used	Section 5.4, Figure 5.24
12	Deluge script on form submission (DS-01 and validation scripts)	Used	Sections 5.5, 6.3, 6.6; Appendix C
13	Deluge validation with alert and cancel submit	Used	Figures 6.1, 6.2, 6.4 – 6.6
14	Deluge fetch records and for each loop	Used	Patient Data Validation (Figure 6.2)
15	Standalone Deluge functions	Not used	Section 5.5
16	Form workflows (4)	Used	Section 5.6, Table 5.6
17	Custom schedule (Appointment Reminder Scheduler)	Used – not verified	Figure 5.28; reminder execution not recorded
18	Approvals, Blueprints, Batch workflows, Payments	Not used	Section 5.6
19	E-mail notification (Appointment Confirmation Email)	Used	Section 8.5, Figure 8.5
20	Connection to Zoho Calendar and createEvent	Used – not verified	Section 5.7; calendar-event creation not verified
21	invokeurl, webhooks, Creator REST API	Not used	Section 5.7
22	Role hierarchy (CEO → Admin → Doctor, Receptionist, Accountant)	Used	Section 5.8, Figure 5.33
23	Permission sets (Accountant, Admin, Doctor, Receptionist)	Used	Section 5.8, Figure 5.32
24	Data sharing rules (2)	Used	Section 5.8, Table 5.9
25	Sign-in test with a user of each role	Not used	Not recorded (Section 6.1, TC-11)

S.No.	Zoho Creator feature	Status	Evidence / remark
26	Stage and Production environments (version 2.0)	Used	Section 7.1, Figure 7.1
27	Zoho Creator mobile app / dedicated mobile configuration	Not used	Section 5.11 (preview toggles only)
28	Patient portal	Not used	Section 5.11; future enhancement
29	Zia / AI features of the platform	Not used	Section 5.9
30	Zoho Analytics	Not used	Section 5.9; future enhancement
31	Audit trail, change history, backup/export configuration	Not used	Not evidenced (Section 7.4)
32	Pro-code extension (Catalyst, JavaScript widgets)	Not used	Section 5.10

APPENDIX B
COURSE OUTCOME MAPPING

Table B.1 maps the learning areas of the course CB23F35 – Low-Code Application Development using Zoho Creator to the chapters of this report. The learning areas are stated in the words of the report; the column “CO No.” is left for the course outcome numbers of the syllabus, which are to be entered by the student or the course faculty.

Table B.1: Mapping of Course Outcomes to Report Chapters

S.No.	Learning area	Demonstrated through	Report chapters / sections	CO No.
1	Low-code concepts and the case for low-code over traditional development	Justification for the approach, comparison table, trends, spectrum reflection	1.5, 2.3 – 2.5, 8.5	
2	Requirement analysis and data modelling	Stakeholders, functional and non-functional requirements, ER diagram, field specifications	Chapter 3; 4.2, 4.3	
3	Building forms, reports, pages and dashboards	Five forms, ten report views, Clinic Dashboard	5.2 – 5.4	
4	Automation with Deluge scripts and workflows	DS-01 calendar script, validation scripts, four form workflows, one schedule	5.5, 5.6, 6.3, 6.6; Appendix C	
5	Integration with other services	Zoho Calendar connection and createEvent call	4.4, 5.7	
6	Security, roles and governance	Role–permission matrix, permission sets, role hierarchy, data sharing rules	4.6, 5.8, 7.3, 7.4	
7	Testing, debugging and quality evidence	Test cases, defect log, validation evidence, acceptance plan	Chapter 6	
8	Deployment and environments	Stage and Production publishing, deployment checklist, rollback plan	7.1, Table 7.1	
9	Responsible use of AI and honest reporting	AI assistance statement, limitations, unverified items	5.9, 8.4, 9.1	
10	Reflection and future scope	Learning outcomes, future enhancements	9.2, 9.3	

APPENDIX C
COMPLETE DELUGE SOURCE CODE

This appendix gives the Deluge code of the application as it appears in the Deluge editor. DS-01 is the script described in Section 5.5. The four validation scripts DS-02 to DS-05 (Figures 6.2, 6.4, 6.5 and 6.6) run on “Validations on form submission”; their identifiers are assigned here for reference. The scripts for the Doctor and Appointment forms continue beyond the last line visible in their screenshots, and only the visible lines are reproduced.

Code C.1 – DS-01: Calendar event script

Appointments → Created → Successful form submission → Create Calendar Event

```
// DS-01 | Appointments | Successful form submission | Create Calendar Event
eventDetails = Map();
eventDetails.put("estatus","added");
eventDetails.put("title","Patient Appointment");
dateText = input.Appointment_Date.toString("dd-MMM-yyyy");
timeText = input.Appointment_Time.toString();
appointmentDateTime = (dateText + " " + timeText).toDateTime();
endDateTime = appointmentDateTime.addMinutes(60);
dateTime = Map();
dateTime.put("timezone","Asia/Kolkata");
dateTime.put("start",appointmentDateTime.toString("yyyyMMdd'T'HHmmss") + "+0530");
dateTime.put("end",endDateTime.toString("yyyyMMdd'T'HHmmss") + "+0530");
eventDetails.put("dateandtime",dateTime);
event = Map();
event.put("eventdata",eventDetails);
response = zoho.calendar.createEvent("a5dea53efcc54b709ba293ac40d9bf12",event,"zoho_calendar_connection");
info appointmentDateTime;
info response;
```

Code C.2 – DS-02: Patient Data Validation

Patients → Created or Edited → Validations on form submission

```
// Validate Full Name
if(Full_Name.first_name.isBlank() && Full_Name.last_name.isBlank())
{
    alert "Full Name is required";
    cancel submit;
}
// Validate Phone Number
if(input.Phone_Number1.isBlank())
{
    alert "Phone Number is required";
    cancel submit;
}
// Check duplicate Phone Number
phoneRecords = Patients[Phone_Number1 == input.Phone_Number1];
if(phoneRecords.count() > 0)
{
    for each  rec in phoneRecords
    {
        if(input.ID == null || rec.ID != input.ID)
        {
            alert "Phone Number already exists";
            cancel submit;
        }
    }
}
// No Email Validation
```

Code C.3 – DS-03: Billing Data Validation

Billing → Created or Edited → Validations on form submission

```
// Validate Total_Amount is greater than 0
if(input.Total_Amount <= 0)
{
    alert "Total Amount must be greater than 0.";
    cancel submit;
}
// Validate Payment_Status is not empty
if(input.Payment_Status.isNull() || input.Payment_Status.isEmpty())
{
    alert "Payment Status is required.";
    cancel submit;
}
// Validate Appointments lookup is not empty
if(input.Appointments.isNull())
{
    alert "Appointment selection is required.";
    cancel submit;
}
```

Code C.4 – DS-04: Doctor Data Validation

Doctors → Created or Edited → Validations on form submission

```
// Validate doctor_name is not empty
if(input.doctor_name.isBlank())
{
    alert "Doctor Name cannot be empty";
    cancel submit;
}
// Validate Email format
if(!input.Email.isNull() && !input.Email.isBlank())
{
    emailPattern = "^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\\.[a-zA-Z]{2,}$";
    if(!input.Email.matches(emailPattern))
    {
        alert "Please enter a valid Email address";
        cancel submit;
    }
}
// Validate Phone_Number1 is not empty
if(input.Phone_Number1.isBlank())
{
    alert "Phone Number cannot be empty";
    cancel submit;
}
// Validate Specialization is not empty
if(input.Specialization.isBlank())
{
    alert "Specialization cannot be empty";
    ...   // remaining lines not visible in the screenshot
```

Code C.5 – DS-05: Appointment Data Validation

Appointments → Created or Edited → Validations on form submission

```
// Validate Patients lookup field
if(input.Patients.isNull())
{
    alert "Patient selection is required";
    cancel submit;
}
// Validate Doctors lookup field
if(input.Doctors.isNull())
{
    alert "Doctor selection is required";
    cancel submit;
}
// Validate Appointment_Date is not in the past
if(input.Appointment_Date < zoho.currentdate)
{
    alert "Appointment date cannot be in the past";
    cancel submit;
}
// Validate Appointment_Time is not empty
if(input.Appointment_Time.isNull())
{
    alert "Appointment time is required";
    cancel submit;
}
// Validate Status is one of the allowed options
allowedStatuses = {"Scheduled","Completed","Cancelled"};
...   // remaining lines not visible in the screenshot
```

APPENDIX D
USER MANUAL

This manual describes how the clinic staff use the Clinic Appointment & Patient Management System. The tasks are written for the Receptionist, Doctor and Administrator roles of Table 4.7. Sign-in with users of each role has not yet been tested (Chapter 6), so the role-wise visibility described here is the intended behaviour of the permission sets.

D.1 Opening the application

- Open the live application at https://creatorapp.zoho.in/renechristine.a.2024.csbs324/clinic-appointment-patient-management-system and sign in with the account given by the administrator.
- The left menu lists Patients, App Preferences, Doctors, Appointments, Treatments, Billings and Clinic Dashboard (Table 5.1).
- Each section holds a data-entry form and its reports. Click a row in any report to open the full record.

D.2 Receptionist tasks

Register a patient

- Open **Patients → Patients** and enter the first and last name, date of birth, phone number, gender and, if available, e-mail, address, blood group and medical history.
- Click **Submit**. If the phone number already belongs to a saved patient, the alert “Phone Number already exists” appears and nothing is saved; open **All Patients** and use the existing record.

Book an appointment

- Open **Appointments → Appointments**. Select the patient and the doctor from the drop-down lists, enter the appointment date, time and reason for visit, and keep the status **Scheduled**.
- Click **Submit**. The record appears in Today’s Appointments or Upcoming Appointments, a confirmation e-mail is sent and a Zoho Calendar event is requested.

Prepare a bill

- Open **Billings → Billing**, select the patient and the appointment, and enter the consultation fee and medicine charges.
- Check the total amount, keep the payment status **Pending** until the money is received, then change it to **Paid**. Unpaid bills are listed in **Pending Bills**.

D.3 Doctor tasks

- Open **Appointments → Today’s Appointments** or **Upcoming Appointments** to see the visits.
- Open **Treatments → Treatments**, select the appointment and enter the diagnosis, prescription and treatment notes. Earlier diagnoses can be read in **All Treatments**.
- After the consultation, update the appointment status to **Completed** or **Cancelled**.

D.4 Administrator tasks

- Maintain doctors in **Doctors → Doctors**; set Availability_Status to **On Leave** when a doctor cannot be booked.

- Open **Clinic Dashboard** for the summary cards, the Appointments by Status and Appointments by Date charts, and the embedded Upcoming Appointments and Pending Bills reports.
- Manage permission sets, roles and data sharing rules under **Settings → Users**, and publish new versions from **Deploy → Environments**.

D.5 Messages shown by the forms

Table D.1: Validation Messages and What to Do

Form	Message	What to do
Patients	Full Name is required	Enter a first or last name.
Patients	Phone Number is required	Enter the phone number.
Patients	Phone Number already exists	Search All Patients; use the saved record.
Appointments	Patient selection is required / Doctor selection is required	Select the patient and the doctor.
Appointments	Appointment date cannot be in the past	Choose today’s date or a later date.
Appointments	Appointment time is required	Enter the time of the visit.
Billing	Total Amount must be greater than 0.	Enter the fee and medicine charges.
Billing	Payment Status is required. / Appointment selection is required.	Select the payment status and the appointment.
Doctors	Doctor Name cannot be empty; Phone Number cannot be empty; Specialization cannot be empty	Complete the missing field.
Doctors	Please enter a valid Email address	Correct the e-mail format.

Known points for users: the application is on a trial edition and holds sample data only; a bill can still be saved without a patient unless the lookup is completed (BUG-04); and the clinic should not enter real patient records until the controls in Section 7.3 are added.